What Bars Cannot See: Order Flow, Fee-Free Volume, and OHLCV Microstructure Proxies

Seyed Abolfazl Hosseini*

November 2026

Abstract

Microstructure measures built from open-high-low-close-volume (OHLCV) bars, such as bulk volume classification (BVC), Kyle's lambda, the Amihud ratio, VPIN and high-low spread estimators, are used whenever trade-level data are unavailable. Binance reports the aggressor-side (taker) volume of every bar, so the signed order flow these measures target is observed. Using hourly and one-minute bars for twenty cryptocurrencies from 2022 to 2026, we establish three results. First, bars measure volatility almost perfectly but order flow poorly: in nineteen of the twenty coins no function of the bar's own return explains more than about a quarter of the variance of hourly aggressor imbalance (0.25 for bitcoin), and BVC is close to this bound. Second, a natural experiment shows that volume-based measures mistake volume for liquidity. When Binance removed all fees on bitcoin pairs for eight months, trades on BTC/USDT rose by a factor of 5.3 relative to nineteen control coins and fell back at the hour fees returned, and 93% of the additional volume was offset by opposite aggressor flow within the same minute. The Amihud ratio and BVC-based Kyle's lambda fell by 69% and 33%, signalling a large gain in liquidity, while the price impact of a dollar of observed net flow did not fall (its change, 28%, is not significant); the same balanced volume then moved from pair to pair as the promotion moved. Third, the part of order flow that bars cannot see is the part that predicts prices: aggressor imbalance is followed by a reversal of about a basis point over two hours, driven by market-wide flow, which observed flow detects out of sample and hourly BVC does not, but which is far too small to trade on. Bar-based flow and liquidity measures are better read as measures of volatility and volume.

Keywords: order flow, bulk volume classification, liquidity measures, trading fees, natural experiment, wash trading, cryptocurrency, return predictability
JEL: G12, G14, G18, C58

*Department of Industrial Engineering, Ferdowsi University of Mashhad. Email: hosseini.systems@gmail.com.
Code and data access: https://github.com/a0x0h/what-bars-cannot-see.

1 Introduction

Microstructure theory ties short-horizon price changes to the arrival of signed order flow. Informed traders reveal information through the direction of their trades (Kyle, 1985; Glosten and Milgrom, 1985), liquidity providers set prices to manage inventory and adverse selection, and order flow has a price impact that is partly permanent and partly transient (Bouchaud et al., 2018). Testing these ideas requires knowing which side initiated each trade. Because such data are expensive or unavailable for many markets and periods, a large applied literature reconstructs microstructure quantities from open-high-low-close-volume (OHLCV) bars. Spreads are estimated from high and low prices (Corwin and Schultz, 2012; Abdi and Ranaldo, 2017), illiquidity from returns and volume (Amihud, 2002), and signed flow by bulk volume classification (BVC), which splits each bar's volume between buyers and sellers according to the standardised price change (Easley et al., 2012, 2016). These variables are increasingly used as features in machine-learning forecasts of returns, including in cryptocurrency markets (Easley et al., 2021, 2024).

Three questions are rarely asked together. How much of the underlying quantity do bar-based proxies capture? Can they tell liquidity from volume when the composition of trading changes? And does the part of order flow they miss matter for prices? We answer them in a setting where the ground truth is observed. Binance, the largest cryptocurrency spot exchange, publishes for every bar the volume executed by buyer-initiated (taker) orders, so the signed order flow of every hour and every minute is known. We verify on one month of trade-level data that the bar-level taker volume reproduces trade-level aggressor flow exactly (correlation 1.0000 over 720 hours). We use hourly and one-minute bars for twenty coins from January 2022 to October 2026, about 42,000 hours and 2.5 million minutes per coin, and an eight-month period in which Binance charged no fees on bitcoin pairs as a natural experiment.

Measurement. We score each proxy against its observed counterpart. Range-based volatility from hourly bars has a correlation of 0.99 with realised volatility from one-minute returns for bitcoin and at least 0.93 for every coin, so bars measure volatility well. Signed flow is another matter. For bitcoin, hourly BVC imbalance has a correlation of 0.49 with observed taker imbalance and the correct sign in 72% of hours. The reason is not the functional form. BVC and the tick rule are functions of the bar's standardised return, and we show nonparametrically that *no* such function explains more than 0.25 of the variance of hourly taker imbalance (0.34 on one-minute bars), even in sample. Across the twenty coins the bound lies between 0.12 and 0.27, lowest for the least traded coins, with one exception (TRX, 0.46). BVC is close to the bound everywhere, so better calibration cannot help much, and its scale drifts over time. Most aggressor flow is absorbed without moving the bar's price. Kyle's lambda from BVC-signed volume inherits the error, and spread estimators measure volatility: on a market whose minimum (one-tick) spread is about 0.002 basis points, the Corwin-Schultz estimate has a median of 15.8 basis points, the extreme case of the bias described by Jahan-Parvar and Zikes (2023).

Volume is not liquidity. A change in trading fees separates volume from order flow, because a fee is a cost per unit of volume, not per unit of information. From 14:00 UTC on 8 July 2022 to 00:00 UTC on 22 March 2023 Binance charged no fees on BTC/USDT and twelve other bitcoin

pairs, while the USDT pairs of other coins paid standard fees. We compare BTC/USDT with nineteen control coins in event windows around both switches, and use permutation tests across coins and over time because there is a single treated unit (Conley and Taber, 2011; Abadie et al., 2010). Removing fees multiplied the number of trades by 5.3 relative to the controls, and the effect started in the hour fees were removed and reversed in the hour they returned. Almost all of the new volume was two-sided: 93% of the excess dollar volume was offset by opposite aggressor flow within the same minute, and in trade-level data the share of volume offset within the same second doubled. The Amihud ratio fell by 69% and Kyle's lambda from BVC-signed flow by 33%, as if the market had become much more liquid. Price impact per dollar of observed net flow did not fall (it changed by 28%, which is not significant), and order-book evidence shows that spreads widened and depth fell (Galati, 2024). VPIN computed with BVC barely moved, while VPIN computed from observed flow fell by 66%. When the zero-fee promotion was narrowed to BTC/TUSD and later moved to BTC/FDUSD, the same high-volume, balanced-flow signature moved with it. Bar-based measures cannot distinguish volume that moves prices from volume that is offset within the minute, and the latter responds to fees and incentives rather than to information.

Information. Observed flow is informative about future prices, in a specific way. For bitcoin, a one standard deviation increase in hourly taker imbalance is followed by a price decline of 1.3 basis points over the next two hours ($t = -4.0$) that fades within a day. The reversal survives controls for lagged returns, momentum, volume and volatility, is of similar size in both halves of the sample and under standard fees, and is unchanged when the bar's own return is held fixed, while the return itself has no predictive power. The information therefore resides in the part of flow that leaves no trace in the hourly bar, which hourly BVC cannot recover and, consistently, does not predict; BVC on one-minute bars recovers part of it, and observed flow subsumes it. Across the twenty coins the response is negative for 17, and the pooled estimate is -0.65 basis points ($t = -3.0$ with Driscoll-Kraay standard errors). The reversal follows market-wide imbalance; coin-specific imbalance does not predict coin-specific returns.

Prediction and economic value. The effect is statistically detectable but economically tiny. A one-variable model with observed imbalance has a small but significant out-of-sample R^2 for bitcoin (0.021% at two hours, Clark-West $p = 0.0001$) and breaks even at about one basis point per unit of turnover. In a monthly walk-forward evaluation with purged training data, tuning inside each training window and automated leakage tests, none of 8 elastic-net and gradient-boosting specifications built on technical indicators, microstructure proxies, observed flow or realised measures beats the naive zero forecast over 53 test months, and the best of 16 trading rules has a deflated Sharpe ratio (Bailey and López de Prado, 2014) of 0.70.

Contributions. First, we provide what is, to our knowledge, the broadest direct validation of bar-based order-flow, price-impact, volatility and spread proxies against exchange-reported aggressor flow, twenty coins over almost five years, with an information ceiling that applies to every classifier based on the bar's return. Second, we use a fee change with untreated controls, known switch hours in both directions and the migration of the promotion across pairs to show what kind of volume fee-free trading creates and how bar-based liquidity and toxicity measures misread it. Two recent

studies use the same event to study spreads, depth and the informativeness of order flow (Galati, 2024; Bozzetto et al., 2026); our question is different: what the event does to the measures that can be computed from bars, which is all that much empirical work has. Third, we show that where order flow predicts intraday returns, its predictive content lies in the component that bar-based proxies cannot see, and that it is market-wide. Fourth, we release a fully reproducible pipeline with leakage tests, so that every result, including the negative forecasting results, can be checked rather than taken on trust.

The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the data and the institutional setting, Section 4 defines the measures, and Section 5 validates the proxies on bitcoin and on twenty coins. Section 6 studies the zero-fee natural experiment. Section 7 asks what observed and proxy flow tell us about future prices, Section 8 reports the out-of-sample forecasting and trading results, Section 9 the robustness checks, Section 10 discusses limitations and Section 11 concludes.

2 Related literature

Do low-frequency liquidity measures measure liquidity? A long line of work benchmarks spread and price-impact proxies built from daily data against estimates from intraday trades and quotes. Goyenko et al. (2009) establish the approach for U.S. equities; Corwin and Schultz (2012) and Abdi and Ranaldo (2017) propose the high-low and close-high-low estimators that now dominate applied work. Jahan-Parvar and Zikes (2023) show that several of these estimators carry a volatility-induced bias that dominates when the true spread is small relative to volatility, and that the bias has grown as spreads have narrowed. In cryptocurrency markets, Brauneis et al. (2021) compare low-frequency measures with order-book benchmarks for bitcoin and ether on three exchanges and find that the Corwin-Schultz and Abdi-Ranaldo estimators track time variation in liquidity best, while Amihud and Kyle-Obizhaeva measures capture levels. We add to this literature in three ways. First, we score proxies of *signed order flow*, not only of spreads, against an exchange-reported ground truth for twenty coins over almost five years. Second, we use a fee change as a controlled shock to the composition of volume, which tests whether volume-based liquidity measures track liquidity or merely volume. Third, we ask whether the measurement error matters for the downstream use that motivates many applications, namely short-horizon return forecasting.

Trade classification and flow toxicity. When aggressor flags are unavailable, trades are signed by the tick rule or by Lee and Ready (1991). Easley et al. (2012, 2016) propose bulk volume classification (BVC), which assigns a fraction $\Phi(\Delta p/\sigma)$ of each bar's volume to buyers, and argue that it captures the information content of order flow better than trade-level rules. Evidence on its accuracy is mixed. Chakrabarty et al. (2015) find that BVC is less accurate than the tick rule and Lee-Ready for individual equities, Panayides et al. (2019) find that it identifies aggressors less accurately but is more closely related to informed trading and improves after calibration, and Andersen and Bondarenko (2014) question whether VPIN built on BVC predicts toxicity events. These studies use equity or futures data over comparatively short samples. We exploit the fact that Binance reports the taker-buy volume of every bar to evaluate BVC on hourly and one-minute bars

over about 42,000 hours for each of twenty coins, and to bound how much of the true flow any return-based classifier can recover.

Trading fees, fee-free volume and wash trading. Theory predicts that trading fees shape quotes, depth and the mix of traders (Colliard and Foucault, 2012; Foucault et al., 2013), and natural experiments in equity markets show that changes in fees and transaction taxes move spreads, depth, volume and market composition (Malinova and Park, 2015; Colliard and Hoffmann, 2017). In cryptocurrency markets, Brauneis et al. (2022) study the introduction of maker fees on Coinbase Pro and find wider quoted spreads, less depth and fewer but larger trades. Two recent studies use the Binance promotion that we also exploit. Galati (2024) shows with order-book data that removing maker and taker fees led market makers to widen spreads and reduce depth, so that total transaction costs rose despite the waiver, and raises concerns about trading conduct; Bozzetto et al. (2026) use the return of fees in March 2023 and conclude that zero fees made order flow less informative about prices. We use the same event for a different purpose, as a controlled shock to the composition of volume that reveals what bar-based measures respond to, with nineteen untreated coins as controls and the restoration of fees as a second test. The episode also connects to the literature on fake volume in cryptocurrency markets, which detects wash trading with statistical tests (Cong et al., 2023; Amiram et al., 2025) and with leaked account-level records (Aloosh and Li, 2024). Aggressor flags do not identify the parties to a trade, so we cannot label individual trades as wash trades. We show instead that the volume created by zero fees was two-sided within the minute and within the second, and that bar-based measures read it as liquidity.

Order flow and short-horizon returns. Order imbalance predicts short-horizon index returns (Chordia et al., 2002), order-flow imbalance at the top of the book explains contemporaneous price changes (Cont et al., 2014), and deep-learning models extract predictive signals from order-book flow at horizons of seconds to minutes (Kolm et al., 2023). For cryptocurrencies, Makarov and Schoar (2020) document that signed order flow explains a large share of bitcoin returns across exchanges, Aleti and Mizrach (2021) compare spot and futures microstructure, and Easley et al. (2024), following Easley et al. (2021), report that microstructure measures such as the Roll measure and VPIN predict future crypto market dynamics in a machine-learning framework. Time-series predictability at horizons of days to weeks is documented by Liu and Tsyvinski (2021). Our study differs from this work in focusing on the intraday horizon at which microstructure effects should be strongest, in measuring predictive gains strictly out of sample with tests designed for nested and overlapping forecasts (Clark and West, 2007; Campbell and Thompson, 2008), and in pricing the resulting signals at realistic transaction costs. Because a single dataset invites specification search, we fix the design before estimation, report every feature set and model we estimate, and interpret Sharpe ratios with the caution recommended by Bailey and López de Prado (2014) and Harvey et al. (2016).

3 Data and institutional setting

Source. We use Binance spot klines for BTC/USDT and ETH/USDT at one-hour and one-minute frequency, downloaded from the exchange's public data archive (`data.binance.vision`). Each bar

reports open, high, low and close prices, base-asset volume V_t, the number of trades, and the base volume of trades whose taker was the buyer, B_t. The observed signed volume of a bar is $2B_t - V_t$ and the observed imbalance is $(2B_t - V_t)/V_t \in [-1, 1]$. Timestamps are in UTC. The archive switched from millisecond to microsecond resolution in January 2025, and we normalise all timestamps. We remove bars with non-positive prices or a close outside the high-low range and do not fill missing bars; rolling statistics that would span a gap are set to missing. The sample runs from 1 January 2022 to 31 October 2026 and contains one missing hour per asset.

Ground truth. The validity of the exercise rests on the taker volume being the true aggressor-side volume. We verify this for September 2026 by downloading every aggregated trade of BTC/USDT, signing each by its buyer-maker flag, and summing to the hour. The hourly imbalance computed from trades and from klines has a correlation of 1.0000 and a mean absolute difference of 0.00011 over 720 hours.

Twenty coins and other bitcoin pairs. To test whether the results generalise, and to build a control group for the natural experiment of Section 6, we download the same hourly and one-minute bars for the USDT pairs of eighteen further coins that were listed on Binance throughout the sample: BNB, XRP, ADA, SOL, DOGE, LTC, TRX, LINK, DOT, AVAX, BCH, ATOM, ETC, XLM, UNI, FIL, NEAR and ALGO. With BTC and ETH they span median hourly dollar volumes from about \$0.3 million to \$60 million. For Section 6 we also use hourly bars of bitcoin quoted in other stablecoins (BTC/BUSD, BTC/TUSD, BTC/FDUSD and BTC/USDC) and every aggregated trade of BTC/USDT and ETH/USDT on the two days before and the two days after each fee switch.

Tick size and fees. BTC/USDT and ETH/USDT trade with a tick of 0.01 USDT, about 0.002 basis points for BTC and 0.043 for ETH at median sample prices, which is the minimum possible quoted spread. The standard spot fee for regular accounts is 10 basis points per side, reduced for high-volume tiers and fee discounts. From 14:00 UTC on 8 July 2022 to 00:00 UTC on 22 March 2023 Binance charged no fees on BTC/USDT and twelve other bitcoin pairs, while the USDT pairs of all other coins kept their fees. The median number of trades per hour on BTC/USDT rose from 41,062 before the promotion to 221,938 during it and fell to 29,443 after it ended, with median hourly dollar volume moving from \$60 million to \$178 million and back to \$34 million. On ETH/USDT the median trade count was 28,242, 27,933 and 15,322 over the same windows. Section 6 studies this episode as a natural experiment; elsewhere we report results with and without the promotion.

Descriptive statistics. Table 1 summarises the two series. Hourly log returns have a standard deviation of about 54 basis points for BTC and 71 for ETH, with fat tails. Observed taker imbalance has a mean close to zero, a standard deviation of 0.13 and modest persistence. Its correlation with the same hour's return is only 0.39 for both assets, an early sign that prices reveal only part of the flow.

Table 1: Descriptive statistics, hourly bars

Asset	Start	End	Hours	Missing	$\sigma(r_{1h})$	$\sigma(r_{4h})$	Kurt(r_{1h})	Med. $vol (M)	Med. trades	Mean imb.	SD imb.	AC(1) imb.	Corr(imb., r_{1h})
BTC/USDT	2022-01-01	2026-10-31	42,359	1	53.5	105.8	13.7	60.3	85,699	−0.021	0.130	0.18	0.39
ETH/USDT	2022-01-01	2026-10-31	42,359	1	71.2	142.3	16.3	31.1	48,442	−0.009	0.126	0.21	0.39

$\sigma(r)$: standard deviation of hourly and four-hour log returns in basis points. $vol: quote (USDT) volume per hour. Imb.: observed taker imbalance $(2B_t - V_t)/V_t$; AC(1) is its first-order autocorrelation.

4 Measures

All measures at bar t are functions of bars up to and including t; Appendix Table 16 lists definitions and windows. We group them by the quantity they target.

Spread and price impact (proxy). The two-bar Corwin and Schultz (2012) high-low estimator averaged over 24 hours; a causal version of the Abdi and Ranaldo (2017) close-high-low estimator that uses bars $t-1$ and t only; the Roll (1984) estimator over 48 hours; the Amihud (2002) ratio; and a square-root impact measure $|r_t|/\sqrt{V_t/\bar{V}}$ (Bouchaud et al., 2018).

Order flow (proxy). Following Easley et al. (2016), the buy fraction of bar t is $\Phi(r_t/\hat{\sigma}_t)$, where $\hat{\sigma}_t$ is the trailing 168-hour standard deviation of hourly returns, so the BVC signed volume is $(2\Phi(r_t/\hat{\sigma}_t) - 1)V_t$. From it we build imbalance at one and 24 hours, its acceleration (6-hour minus 24-hour imbalance), a VPIN-style toxicity measure (the 24-hour sum of absolute signed volume over total volume), and Kyle's lambda from a rolling 48-hour regression of returns on signed dollar volume.

Observed flow and trade activity. The same five constructions applied to observed signed volume $2B_t - V_t$, which gives matched pairs of proxy and observed features; standardised trade counts and average trade size; and the taker imbalance of the last 15 minutes of the hour.

Volatility and intrahour measures. Range-based estimators of Garman and Klass (1980), Rogers and Satchell (1991) and Yang and Zhang (2000) over 24 hours, the ratio of 24-hour to 168-hour volatility and volatility of volatility. From one-minute bars of the same hour we compute realised variance, the jump share $\max(RV - BV, 0)/RV$ with bipower variation BV (Barndorff-Nielsen and Shephard, 2004), and realised skewness (Amaya et al., 2015).

Efficiency, bar structure and activity. Efficiency ratio, return autocorrelation, variance ratio, candle body and wick shares, and standardised volume.

Technical controls. Momentum over 1, 4, 24 and 168 hours, RSI(14), MACD histogram, Bollinger %B, gaps to the 24- and 168-hour moving averages, and ATR(14).

Feature sets. TI contains the 10 technical controls; MS_{proxy} the 23 OHLCV-only microstructure measures; MS adds observed flow, trade activity and one-minute measures (35 features); ALL is MS plus TI (45 features). Flow_{BVC} and $\text{Flow}_{\text{taker}}$ are the five matched flow constructions from BVC

and from observed volume. The first four sets are nested, which the incremental tests exploit; the two flow sets differ only in the input.

5 Do bar-based proxies measure what they claim?

This section scores the OHLCV-only measures against the quantities they are meant to capture. Observed signed flow is the taker imbalance $(2B_t - V_t)/V_t$; the benchmark for volatility is realised variance from one-minute returns. Because quoted spreads are not in the public archive, we evaluate the spread estimators indirectly, against the tick size and against volatility. Table 2 collects the main statistics for BTC/USDT; ETH/USDT results are in Appendix Table 28.

Signed order flow. Hourly BVC imbalance has a correlation of 0.49 with observed taker imbalance and the same sign in 72% of hours. Aggregating to 24 hours barely changes the correlation (0.51), so the error is not high-frequency noise that averages out. Applying BVC to one-minute bars and summing within the hour raises the correlation to 0.60, the largest gain available from the data we use. The choice of the volatility window (24 hours to 30 days) or of a heavy-tailed Student-t CDF moves the correlation by at most two hundredths (Appendix Table 18). The tick rule, which signs the whole bar by its return, is worse on hourly bars but nearly as good as BVC on one-minute bars, consistent with Chakrabarty et al. (2015) and Panayides et al. (2019).

An information ceiling for return-based classification. BVC, the tick rule and their variants are all functions of the standardised bar return $z_t = r_t/\hat{\sigma}_t$. Their accuracy is therefore bounded by how much of the observed imbalance is explained by $\mathbb{E}[\text{imb}_t \mid z_t]$, whatever the functional form. We estimate this conditional mean nonparametrically with 50 quantile bins of z_t. As an upper bound we compute its in-sample R^2; with 50 bins and thousands of hours per period the overfitting bias is negligible. Over the full sample the bound is 0.25 on hourly bars and 0.34 on one-minute bars, and in no calendar year does the hourly bound exceed 0.33 (Appendix Table 19). The squared correlation of BVC with observed imbalance is 0.24, almost exactly at the bound: the shape of the BVC map is close to optimal, and no other function of the bar return would do materially better. Fitted on the first half of the sample and evaluated on the second, the out-of-sample ceiling is 0.21, and rescaled BVC attains 0.21. Fitted on the second half and evaluated on the first, however, it falls to 0.09 for BTC (0.23 for ETH), because the scale of the relation drifts: the slope of observed imbalance on BVC rises from 0.065 in 2022 to 0.169 in 2026. A classifier calibrated in one period is therefore miscalibrated in another, so out-of-sample accuracy is lower than the bound.

Figure 1 shows the mechanism. The BVC map $2\Phi(z) - 1$ has the right shape, but observed imbalance moves far less than the map implies (the slope fitted on the first half of the sample is 0.09 on hourly bars), and its dispersion at any given return is wide. Two conclusions follow. First, the shape of the classifier is not the problem; recalibration, which Panayides et al. (2019) recommend, fixes the scale but must be repeated as the scale drifts. Second, about three-quarters of the hourly variation in aggressor imbalance leaves no trace in the bar's own price change. This flow is absorbed by passive liquidity, offset within the hour, or reflected in prices only later, and no classifier based on the hourly bar can recover it. Part of it is visible in the one-minute price path, which is why BVC on one-minute bars does better.

Table 2: Bar-based proxies against observed quantities, BTC/USDT

	Pearson	Spearman	Same sign	N
Panel A. Signed order flow: proxy vs. observed taker imbalance				
BVC, 1h bars, per hour	0.489	0.541	0.716	42,316
BVC, 1h bars, 4h sums	0.475	0.524	0.701	42,310
BVC, 1h bars, 24h sums	0.507	0.538	0.693	42,270
BVC, 1m bars summed to 1h	0.603	0.637	0.747	42,359
Tick rule, 1h bars, per hour	0.430	0.389	0.716	42,357
Panel B. Price impact and volatility				
Kyle λ (48h): BVC vs. observed flow	0.260	0.213	–	41,978
log GK vol (1h bars) vs. log RV (1m), 24h	0.987	0.983	–	41,978
Panel C. Information ceiling of return-based classifiers (out-of-sample R^2)				
	BVC raw	BVC rescaled	Any $f(z)$	N
1h bars	−6.783	0.209	0.213	21,158
1m bars	0.116	0.308	0.328	1,270,544
Panel D. Spread estimators (one tick = 0.002 bps)				
	Median (bps)	Mean (bps)	Share = 0	Corr. log RV
Corwin-Schultz (24h mean)	15.8	17.5	0.0%	0.87
Abdi-Ranaldo (24h)	0.0	9.6	50.2%	0.64
Roll (48h)	15.1	19.7	38.4%	0.52

Panel A: correlation and sign agreement between proxy and observed taker imbalance $(2B_t - V_t)/V_t$, per hour and over rolling 4- and 24-hour sums. Panel B: Kyle's lambda from 48-hour rolling regressions on BVC-signed and on observed signed dollar volume; Garman-Klass volatility from hourly bars against realised volatility from one-minute returns. Panel C: out-of-sample R^2 for observed imbalance of raw BVC, of BVC after a linear rescaling, and of the best function of the standardised bar return (50 quantile bins), all fitted on the first half and evaluated on the second. Panel D: spread estimators in basis points, share of hours truncated at zero, and correlation of the log estimate with log realised volatility.

Price impact. Kyle's lambda estimated from BVC-signed dollar volume has a correlation of only 0.26 with the same estimator applied to observed signed volume. Errors in the regressor are amplified in a ratio of a covariance to a variance, so a proxy that captures about a quarter of the flow variance yields an impact estimate that is largely noise.

Volatility. The range-based Garman-Klass estimator from hourly bars has a correlation of 0.99 with realised variance from one-minute returns (both in logs, 24-hour windows). Bar-based volatility measures are close substitutes for realised measures at this horizon.

Spreads. The minimum quoted spread on BTC/USDT is one tick of 0.01 USDT, about 0.002 basis points at the sample's median price. The 24-hour Corwin-Schultz estimate has a median of 15.8 basis points, so taken at face value it would imply an average book about four orders of magnitude wider than one tick, an implausible value for the most liquid crypto pair. Its logarithm has a correlation of 0.87 with log realised volatility. The Abdi-Ranaldo estimate is truncated at zero in 50% of hours and the Roll estimate in 38%. In this market the spread estimators measure volatility, the extreme case of the bias documented by Jahan-Parvar and Zikes (2023). Liquidity conclusions drawn from them at hourly frequency should be read as volatility conclusions.

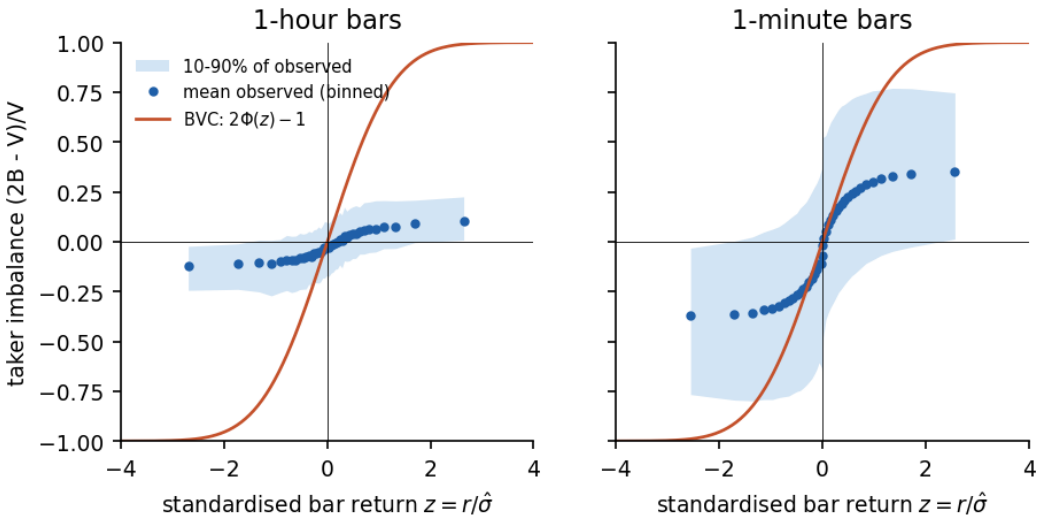


Figure 1: Observed taker imbalance against the standardised bar return. Dots are bin means of observed imbalance in 40 quantile bins of z; the band covers the 10th to 90th percentile within each bin; the line is the BVC mapping $2\Phi(z) - 1$. BTC/USDT, January 2022 to October 2026.

Stability. The BVC correlation varies between 0.48 and 0.56 across calendar years and is essentially the same in low-, medium- and high-volatility regimes at the hourly level (Appendix Table 20). It is 0.60 during Binance's zero-fee promotion on BTC/USDT and 0.51 under standard fees. The promotion changed the flow itself: the standard deviation of hourly taker imbalance on BTC/USDT fell to 0.030 from 0.140 under standard fees, while on ETH/USDT, which kept its fees, it barely changed (0.095 against 0.131). Section 6 studies this episode in detail.

5.1 Twenty coins

Table 3 and Figure 2 repeat the main measurement statistics for all twenty coins. In nineteen of them the hourly bar reveals at most about a quarter of the variation in aggressor flow. The correlation of hourly BVC with observed imbalance has a median of 0.41 and lies between 0.33 and 0.52, and the in-sample information ceiling of return-based classification on hourly bars lies between 0.12 and 0.27. The exception is TRX, with a correlation of 0.68 and a ceiling of 0.46. The ceiling is highest for the most traded coins and lower for smaller coins, whose coarser relative ticks make hourly returns less informative; computed on one-minute bars, BVC is more uniform across coins (correlation between 0.46 and 0.79). In every coin the squared BVC correlation is close to the ceiling, so the BVC map is near the best a return-based classifier can do. Range-based volatility has a correlation between 0.93 and 0.99 with realised volatility, and the median Corwin-Schultz spread is between 4 and 9,996 ticks. That most hourly aggressor imbalance leaves no trace in the bar's price is therefore not a feature of one market but a regularity of this trading environment, and it is strongest where trading is thinnest.

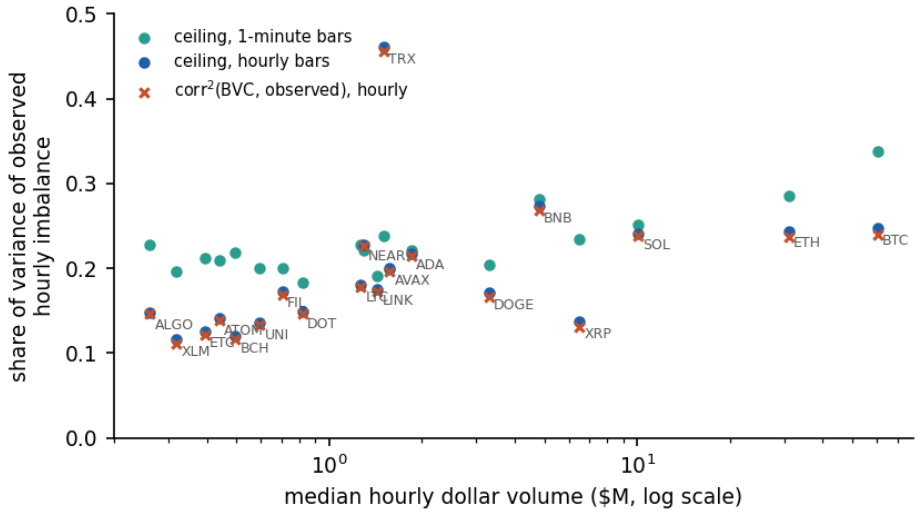


Figure 2: Information ceiling of return-based classification (in-sample R^2 of observed hourly imbalance on 50 bins of the standardised bar return) on hourly and one-minute bars, and squared correlation of hourly BVC with observed imbalance, against median hourly dollar volume. One point per coin, January 2022 to October 2026.

Table 3: Measurement and flow response across twenty coins

Coin	$vol	corr(BVC, observed)		Ceiling		corr(GK,	CS /	2h response to 1 SD flow, bps (t)		SP
	$M/h	1h bars	1m bars	1h	1m	RV)	tick	observed	given z_t	CW p
BTC	60.3	0.49	0.60	0.25	0.34	0.99	9996	−1.30 (−4.0)	−1.41 (−3.7)	<0.001
ETH	31.1	0.49	0.56	0.24	0.29	0.99	478	−0.53 (−1.1)	−0.79 (−1.5)	0.444
SOL	10.1	0.49	0.59	0.24	0.25	0.98	30	−0.81 (−1.3)	0.25 (0.4)	0.187
XRP	6.5	0.36	0.46	0.14	0.23	0.99	16	0.74 (1.4)	1.29 (2.1)	0.030
BNB	4.8	0.52	0.62	0.27	0.28	0.98	10	−0.47 (−1.2)	0.04 (0.1)	0.286
DOGE	3.3	0.41	0.55	0.17	0.20	0.99	25	−0.70 (−1.2)	0.24 (0.4)	0.142
ADA	1.9	0.46	0.62	0.22	0.22	0.98	11	−0.78 (−1.4)	−0.58 (−0.9)	0.452
AVAX	1.6	0.44	0.56	0.20	0.20	0.96	7	−1.57 (−2.5)	−1.79 (−2.6)	0.011
TRX	1.5	0.68	0.79	0.46	0.24	0.94	7	1.66 (6.3)	0.78 (1.4)	<0.001
LINK	1.4	0.41	0.56	0.18	0.19	0.96	20	−0.27 (−0.5)	−0.18 (−0.3)	0.844
NEAR	1.3	0.48	0.61	0.23	0.22	0.98	10	−0.87 (−1.2)	0.07 (0.1)	0.305
LTC	1.3	0.42	0.59	0.18	0.23	0.98	18	−1.37 (−2.7)	−0.60 (−1.1)	0.025
DOT	0.8	0.38	0.56	0.15	0.18	0.94	12	−1.39 (−2.6)	−0.97 (−1.6)	0.052
FIL	0.7	0.41	0.57	0.17	0.20	0.94	11	−0.05 (−0.1)	−0.15 (−0.2)	0.892
UNI	0.6	0.36	0.58	0.14	0.20	0.97	20	−0.91 (−1.4)	−1.05 (−1.5)	0.308
BCH	0.5	0.34	0.56	0.12	0.22	0.97	8	0.10 (0.2)	0.21 (0.3)	0.252
ATOM	0.4	0.37	0.61	0.14	0.21	0.97	14	−1.04 (−2.0)	−0.40 (−0.7)	0.090
ETC	0.4	0.35	0.54	0.13	0.21	0.97	5	−0.63 (−1.2)	−0.35 (−0.6)	0.754
XLM	0.3	0.33	0.54	0.12	0.20	0.93	4	−1.35 (−2.7)	−1.25 (−2.0)	0.020
ALGO	0.3	0.38	0.61	0.15	0.23	0.97	6	−1.39 (−2.4)	−0.93 (−1.5)	0.023
Median	1.4	0.41	0.58	0.17	0.22	0.97	12	−0.80	−0.38	
Min	0.3	0.33	0.46	0.12	0.18	0.93	4	−1.57	−1.79	
Max	60.3	0.68	0.79	0.46	0.34	0.99	9996	1.66	1.29	

Binance USDT pairs, January 2022 to October 2026, sorted by median hourly dollar volume ($vol, $ million). corr(BVC, observed): correlation of BVC imbalance with observed hourly taker imbalance, BVC computed on hourly bars or on one-minute bars and summed to the hour. Ceiling: in-sample information ceiling of return-based classifiers, R^2 of observed imbalance on 50 quantile bins of the standardised bar return, on hourly and one-minute bars. corr(GK, RV): log Garman-Klass volatility from hourly bars against log realised volatility from one-minute returns, 24-hour windows. CS / tick: median 24-hour Corwin-Schultz spread in multiples of the relative tick. 2h response: regression (2) at $k = 2$ hours (basis points per standard deviation of observed imbalance, Newey-West t-statistic), alone and holding the bar return fixed. SP CW p: Clark-West p-value of the single-predictor out-of-sample test with observed imbalance at two hours.

6 A natural experiment: what zero fees did to the bars

The comparisons so far use a sample in which the composition of trading changes slowly. A change in trading fees gives a sharper test of what bar-based measures respond to. A fee is a cost per unit of volume, not per unit of information. Removing it makes a round trip, a purchase and an offsetting sale of the same quantity, nearly free on a pair whose quoted spread is one tick, so it can add volume and trades without adding net order flow. Measures that scale by volume, such as the Amihud ratio and the Kyle's lambda and VPIN built from BVC-signed volume, would then register a change in liquidity or in the toxicity of flow, while measures built from observed net flow would not. Binance's zero-fee promotion lets us run this test with untreated coins as controls, with switch times that are known to the hour, and with the reversal of the treatment as a second test.

6.1 Institutional setting and design

The promotion. On 6 July 2022 Binance announced that from 8 July, until further notice, it would charge neither maker nor taker fees on thirteen spot pairs of bitcoin against stablecoins and fiat currencies, including BTC/USDT and BTC/BUSD (CoinDesk, 2022). Within hours of the start, after a surge of trading at almost unchanged prices, the exchange excluded these pairs from the volume that determines VIP fee tiers, with the stated aim of removing the incentive to wash trade (Decrypt, 2022). Standard fees returned on BTC/USDT and the other pairs on 22 March 2023, except on BTC/TUSD, which stayed free until 7 September 2023 (CoinDesk, 2023; crypto.news, 2023); by then BTC/FDUSD, which opened in August 2023, was the only bitcoin pair with neither fee (crypto.news, 2023). The USDT pairs of other coins paid standard fees throughout.

Timing. The switch times can be read off the data (Figure 3). The number of trades on BTC/USDT jumps between the 13:00 and 14:00 UTC bars of 8 July 2022 and drops at the 00:00 UTC bar of 22 March 2023, while the average control coin does not move. Measured relative to the controls, BTC trade counts in the 24 hours after the first switch were 3.7 times those in the 24 hours before, and 0.22 times after the second switch. Over the same 24-hour comparison at 991 placebo hours in 2024–2026 the largest absolute change in log trade counts is 0.93, against 1.30 and 1.53 at the two switches.

Daily measures. For each coin and day we compute, from one-minute bars, the number of trades, the dollar volume V and the average trade size; the *directional volume* $\sum_m |B_m - S_m|$ and the *two-sided volume*

$$V^{2s} = \sum_m (B_m + S_m - |B_m - S_m|) = \sum_m 2\min(B_m, S_m), \tag{1}$$

where B_m and S_m are the dollar volumes of buyer- and seller-initiated trades in minute m, so that two-sided volume is the part of the day's volume that is offset by opposite aggressor flow within the same minute; the mean absolute hourly taker imbalance; realised volatility; the Amihud ratio from hourly bars; Kyle's lambda as the slope of one-minute returns on signed dollar flow, with flow signed by BVC and, alternatively, observed; VPIN computed from hourly BVC-signed and from observed

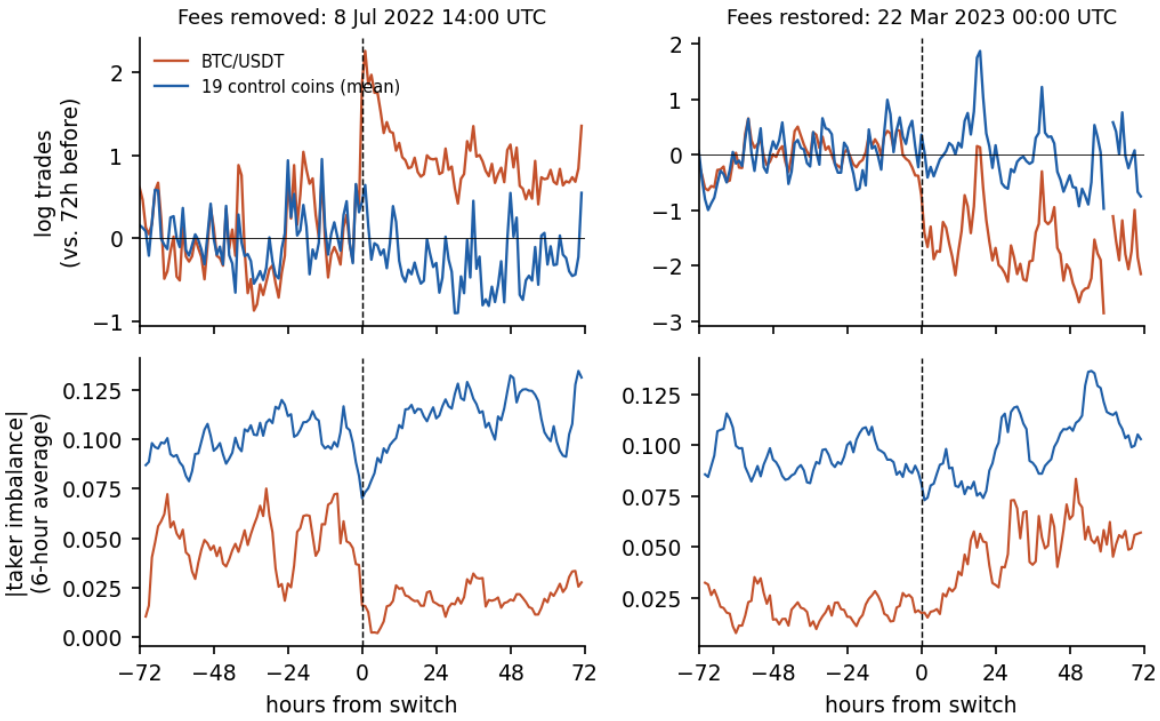


Figure 3: Hourly trades and taker imbalance around the two fee switches. Top: log number of trades relative to the mean over the 72 hours before the switch, for BTC/USDT and for the average of the 19 control coins. Bottom: absolute hourly taker imbalance $|2B_t - V_t|/V_t$, six-hour moving average. The dashed lines mark the first hour without fees (left) and the first hour with fees again (right).

signed volume; the R^2 of one-minute returns on observed flow; the correlation of one-minute BVC imbalance with observed imbalance; and the first-order autocorrelation of one-minute returns.

Design. BTC/USDT is the treated unit and the 19 other USDT pairs of Section 3 are the controls. For each measure let D_d be the value for BTC on day d minus the average over the controls (in logarithms for positive quantities), which removes market-wide movements as day fixed effects would. The effect of a switch is the change in the mean of D_d from the 90 days before to the 90 days after it, estimated by regressing D_d on an indicator for the later window with Newey-West standard errors (30 lags); 8 July 2022, which is split between the two regimes, is left out. With a single treated unit, conventional standard errors can understate uncertainty (Conley and Taber, 2011), so we add two permutation tests. In the *in-space* test each control in turn is treated as if it had been treated, with the remaining controls as its comparison group, and the p-value is the rank of BTC's absolute effect among all 20 coins; the smallest attainable value is 0.05 (Abadie et al., 2010). In the *in-time* test the same statistic is computed for BTC at 123 weekly placebo dates in 2024–2026, when no fee change affected BTC/USDT, and the p-value is the share of placebo effects at least as large in absolute value. The identifying assumption is that, without the fee change, the difference between BTC and the average control would not have shifted at the switch. The restoration of fees provides a second, independent check: a bitcoin-specific shock that happened to coincide with the removal of fees would have to be undone, at the same hour of the day, eight and a half months later.

Table 4: Effects of removing and restoring fees on BTC/USDT

	BTC/USDT, daily median			Fees removed			Fees restored		
Measure	Before	Zero fee	After	Δ (t)	p_{space}	p_{time}	Δ (t)	p_{space}	p_{time}
Panel A. Trading activity									
Trades per day (millions)	1.12	5.74	0.82	1.68 (12.2)	0.050	0.008	−1.68 (−18.1)	0.050	0.008
Dollar volume per day ($bn)	1.78	4.73	1.05	1.21 (5.9)	0.100	0.008	−1.18 (−11.2)	0.050	0.008
Average trade size ($)	1,566	826	1,259	−0.46 (−4.3)	0.100	0.081	0.50 (7.5)	0.050	0.073
Panel B. Direction of the volume (one-minute taker flow)									
Directional volume $\sum_m \lvert B_m - S_m \rvert$ ($bn)	0.44	0.55	0.36	0.43 (3.3)	0.200	0.185	−0.27 (−3.1)	0.350	0.565
Two-sided volume ($bn)	1.35	4.20	0.70	1.43 (6.0)	0.100	0.008	−1.33 (−12.0)	0.050	0.008
Two-sided share of volume	0.74	0.88	0.66	0.16 (6.2)	0.050	0.008	−0.15 (−8.9)	0.050	0.008
Mean |hourly taker imbalance|	0.065	0.023	0.086	−1.03 (−9.3)	0.050	0.008	1.15 (21.3)	0.050	0.008
Panel C. Liquidity and flow measures computed from bars									
Amihud ratio, hourly bars ($\lvert r \rvert$ per $bn)	0.064	0.013	0.052	−1.18 (−6.9)	0.100	0.008	1.52 (21.8)	0.050	0.008
Kyle's λ, BVC-signed flow (bps per $M)	3.81	1.53	3.05	−0.39 (−3.1)	0.150	0.032	0.97 (15.2)	0.050	0.008
VPIN, BVC	0.50	0.45	0.47	−0.10 (−4.7)	0.050	0.008	0.11 (5.1)	0.050	0.008
Panel D. The same quantities from observed taker flow									
Kyle's λ, observed flow (bps per $M)	6.29	5.55	5.90	0.24 (1.4)	0.400	0.435	0.30 (4.1)	0.250	0.290
VPIN, observed flow	0.069	0.023	0.083	−1.08 (−8.8)	0.050	0.008	1.11 (23.1)	0.050	0.008
R^2 of 1m returns on observed flow	0.24	0.41	0.40	0.13 (5.2)	0.050	0.008	−0.06 (−3.8)	0.050	0.081
corr(BVC, observed imbalance), 1m	0.50	0.59	0.61	0.05 (4.1)	0.150	0.008	−0.05 (−6.2)	0.100	0.008
Panel E. Prices									
Realised volatility (% per day)	3.05	2.42	1.76	0.15 (1.4)	0.900	0.742	0.48 (1.9)	0.150	0.169
First-order autocorrelation of 1m returns	0.012	−0.002	0.016	−0.01 (−0.5)	0.850	0.677	0.02 (2.0)	0.250	0.306

Daily medians for BTC/USDT from 1 January to 7 July 2022 (before), 9 July 2022 to 21 March 2023 (zero fee) and 22 March to 6 September 2023 (after). Δ: change in the mean of the BTC-minus-control difference between the 90 days before and the 90 days after the switch, in log points for the measures shown in logs (all except the two-sided share, the R^2, the correlation and the autocorrelation, which are in levels), with the Newey-West t-statistic in parentheses. p_{space}: rank of BTC's absolute effect among the 20 coins when each control is treated in turn as if it had been treated. p_{time}: share of 123 placebo dates in 2024–2026 with an absolute effect at least as large.

6.2 Results

Activity. Table 4 reports the effects. Removing fees multiplied the number of trades on BTC/USDT by 5.3 and its dollar volume by 3.4 relative to the controls. Both effects are larger than at any placebo date. The effect on trades is the largest among the twenty coins; for dollar volume one control, ETC, had a larger change, as its trading surged in July 2022 ahead of Ethereum's switch to proof of stake, which was expected to send miners to the Ethereum Classic network. Both effects were reversed when fees returned (−1.68 and −1.18 log points), and there BTC's changes are the largest of all coins. The average trade became smaller.

Where the volume came from. The new volume had little direction. Two-sided volume rose by a factor of 4.2 and the two-sided share of volume by 0.16. Directional volume rose by a factor of 1.5, but this increase is not significant in either permutation test ($p_{\text{space}} = 0.200$, $p_{\text{time}} = 0.185$), and neither is the decline when fees returned; some directional flow may have moved to Binance from other exchanges, whose fees did not change. Relative to the counterfactual implied by the controls, 93% of the additional dollar volume during the promotion was offset within the same minute by opposite aggressor flow. The mean absolute hourly imbalance fell by 64% and recovered when fees returned. Trade-level data around the switches (Table 5) show the same at a finer scale: the share of BTC/USDT volume offset by opposite aggressor flow within the *same second* was 30% on the two days before the fees were removed and 59% on the two days after, and 53% on the two days before

Table 5: Trade-level composition of volume around the fee switches

Days	Pair	Trades (M/day)	Median trade ($)	Round trips: trades	volume	Two-sided, 1s	Two-sided, 1m
fees (before removal)	BTC/USDT	1.17	250	1.2%	0.1%	30%	82%
fees (before removal)	ETH/USDT	0.63	470	0.5%	0.0%	32%	79%
zero fees (after removal)	BTC/USDT	3.16	204	1.4%	1.3%	59%	91%
zero fees (after removal)	ETH/USDT	0.46	285	3.7%	0.4%	20%	72%
zero fees (before restoration)	BTC/USDT	10.38	181	0.6%	0.1%	53%	92%
zero fees (before restoration)	ETH/USDT	0.80	237	0.2%	0.2%	25%	77%
fees (after restoration)	BTC/USDT	1.37	157	0.3%	0.1%	25%	76%
fees (after restoration)	ETH/USDT	0.68	343	0.3%	0.2%	26%	77%

Averages over two days: 6–7 July 2022 (fees), 9–10 July 2022 (zero fees), 20–21 March 2023 (zero fees) and 23–24 March 2023 (fees). Trades are Binance aggregated trades (all fills of one taker order at one price). Round trips: share of trades, and of dollar volume, that are followed or preceded within one second by a trade of identical quantity with the opposite aggressor side. Two-sided: $1 - \sum_s |B_s - S_s| / \sum_s (B_s + S_s)$ over seconds or minutes s.

they were restored and 25% after; on ETH/USDT it was between 18% and 32% on all eight days. Immediate round trips of identical size, a trade followed within one second by an opposite trade of exactly the same quantity, account for at most 1.4% of BTC volume on any of these days. The fee-free volume was therefore not a simple repetition of matched orders but fast two-sided trading in varying sizes, which is what both high-frequency liquidity provision and wash trading designed to avoid detection would produce (Cong et al., 2023; Aloosh and Li, 2024). Aggressor flags cannot tell the two apart.

What the bars say. Bar-based measures read the episode as a large improvement in liquidity. The Amihud ratio fell by 69% relative to the controls and Kyle's lambda estimated from BVC-signed flow by 33%; both returned towards their earlier levels when fees were restored. VPIN computed with BVC changed by only −9%, so a researcher using it would conclude that the toxicity of order flow was unaffected.

What the observed flow says. Observed flow tells a different story. Price impact per dollar of observed net flow, Kyle's lambda with observed signing, changed by 28% ($t = 1.4$, $p_{\text{space}} = 0.400$, $p_{\text{time}} = 0.435$) when fees were removed and by 35% when they returned ($p_{\text{space}} = 0.250$, $p_{\text{time}} = 0.290$), and realised volatility did not change significantly at either switch. VPIN computed from observed flow, by contrast, fell by 66% and recovered at the restoration: the imbalance of aggressor flow per unit of volume collapsed, because the added volume was balanced. A dollar of net order flow moved prices at least as much as before, so liquidity for a trader with a directional order did not improve, whatever the volume measures say; order-book data indicate that it deteriorated, with wider spreads and less depth (Galati, 2024). The Amihud ratio scales price changes by gross volume and VPIN-BVC scales classified flow by gross volume; neither can tell volume that moves prices from volume that is offset within the minute. The changes in the Amihud ratio are roughly the sum of the changes in observed-flow lambda and in observed VPIN (0.24 and −1.08 log points against −1.18 when fees were removed, 0.30 and 1.11 against 1.52 when they returned): most of the movement reflects the share of net flow in volume, not the cost of price impact.

Calibration and information. The episode also changes the calibration of BVC. Within the promotion the slope of observed imbalance on hourly BVC was 0.038, against 0.142 under standard

Table 6: Bitcoin pairs by fee regime

	Jan–Jul 2022 *fees on all*		Jul 2022–Mar 2023 *13 pairs free*		Mar–Sep 2023 *only TUSD free*		Sep–Dec 2023 *FDUSD free*	
Pair	$M/day	|imb.|	$M/day	|imb.|	$M/day	|imb.|	$M/day	|imb.|
BTC/USDT	1,784	0.072	4,728	0.024	1,055	0.086	1,060	0.091
BTC/BUSD	592	0.080	2,231	0.027	132	0.107	20	0.196
BTC/TUSD	0.9	0.437	0.5	0.368	1,419	0.030	82	0.196
BTC/FDUSD	–	–	–	–	9.9	0.181	1,371	0.025
BTC/USDC	71	0.172	170	0.101	13	0.287	25	0.195

Median daily dollar volume ($ million) and mean volume-weighted absolute hourly taker imbalance of each pair, by fee regime. In the second window all five pairs were free; in the third only BTC/TUSD was; in the fourth BTC/FDUSD was the only pair without fees. BTC/FDUSD opened in August 2023.

fees (Appendix Table 19), so a classifier calibrated in one regime misstates the scale of flow in the other by a factor of about four. Its correlation with observed imbalance changed little (0.05 and −0.05 at the two switches). The R^2 of one-minute returns on observed flow rose by 0.13 when fees were removed and fell by 0.06 when they returned: under zero fees, net taker flow tracked contemporaneous price changes more closely, as it would if fee-free traders, for example arbitrageurs across the thirteen bitcoin pairs, took liquidity in the direction of price moves. For aggressor flow we therefore find no sign that zero fees made order flow less informative about prices, the conclusion Bozzetto et al. (2026) reach for order-book imbalance; the two measures capture different flows.

6.3 The volume follows the fee

The strongest evidence that the two-sided volume was a response to fees, and not to anything specific to bitcoin or to USDT, comes from the other bitcoin pairs (Figure 4 and Table 6). While all thirteen pairs were free, BTC/TUSD traded a median of $0.5 million a day. Once it was the only free pair it traded $1,419 million, and its mean absolute hourly imbalance fell from 0.368 to 0.030, the level BTC/USDT had shown while it was free (0.024). When fees returned on BTC/TUSD in September 2023, the volume and the same low-imbalance signature moved to BTC/FDUSD ($1,371 million a day, imbalance 0.025), while the imbalance of BTC/USDT stayed at its pre-promotion level (0.072 before, 0.086 and 0.091 after). The same pattern, high volume with almost perfectly balanced aggressor flow, appeared three times on three different pairs, each time exactly where trading was free.

6.4 Robustness and implications

Appendix Table 17 repeats the estimates with ETH/USDT as the only control, with 60- and 180-day windows, and with a comparison of the full windows before, during and after the promotion, with and without a linear trend in the BTC-minus-control difference. The effects on activity, on the composition of volume and on the bar-based measures keep their sign in every specification and are significant at the 5% level in all but one case, the fall of BVC-based lambda in the full-window regression with a trend. The change in observed-flow lambda, by contrast, takes both signs across

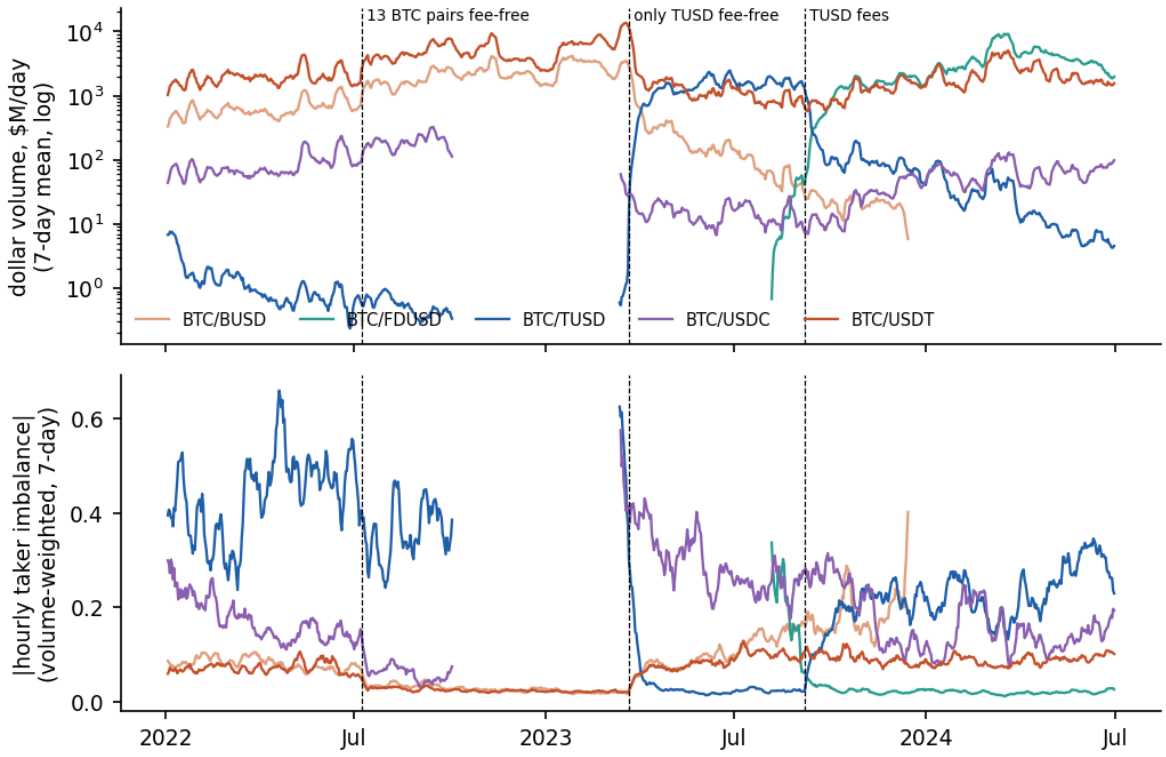


Figure 4: Bitcoin against five stablecoins on Binance: daily dollar volume (top, log scale) and volume-weighted mean absolute hourly taker imbalance (bottom), seven-day moving averages. Dashed lines: zero fees on thirteen bitcoin pairs from 8 July 2022; standard fees on all of them except BTC/TUSD from 22 March 2023; standard fees on BTC/TUSD from 7 September 2023.

specifications. The event-study plots in Appendix Figure 7 show that the effects appear and disappear at the switch dates.

Two implications follow. First, volume-based liquidity measures are not comparable across fee regimes, exchanges or periods with different incentives to trade for its own sake. In cryptocurrency markets, where fee promotions, volume-based fee tiers and wash trading are common (Cong et al., 2023; Amiram et al., 2025), a fall in the Amihud ratio or in BVC-based lambda can mean that more volume is being offset within the minute rather than that liquidity has improved. Second, in the data that zero fees produce, the aggressor imbalance of a bar is diluted by two-sided trading, and the bar's volume says little about how much of it carried direction. Exchange-reported aggressor flow reveals the dilution; bars alone cannot.

7 What does order flow tell us about future prices?

Before turning to out-of-sample forecasting, we ask a simpler question: is the observed flow, or its proxy, related to subsequent returns at all, and if so, through which component? For each horizon $k = 1, \ldots, 24$ hours we estimate

$$\log \frac{C_{t+k}}{C_t} \times 10^4 = a_k + b_k \, \tilde{x}_t + c_k \, \tilde{z}_t + e_{t,k}, \tag{2}$$

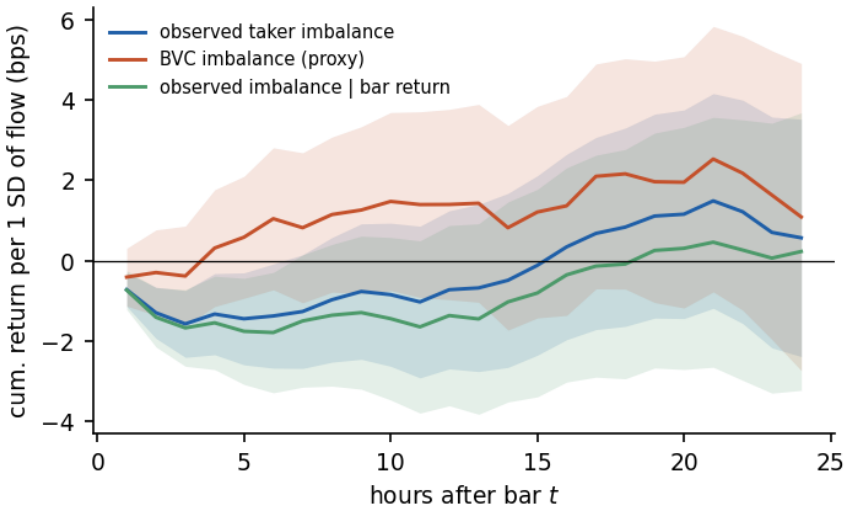


Figure 5: Cumulative log return (basis points) over k hours following a one standard deviation flow shock in hour t, from regression (2), with 95% Newey-West bands. Blue: observed taker imbalance alone. Red: BVC imbalance alone. Green: observed imbalance holding the bar's standardised return fixed. BTC/USDT.

where $\tilde{x}_t$ is a standardised flow measure of bar t (observed taker imbalance or BVC imbalance) and $\tilde{z}_t$ the standardised bar return $r_t/\hat{\sigma}_t$, which is the only input of BVC. The coefficient b_k is the cumulative return, in basis points, that follows a one standard deviation flow shock. When $\tilde{z}_t$ is included, b_k measures the effect of the component of flow that is orthogonal to the bar's own price change, i.e. the part that no return-based classifier can recover. Standard errors are Newey-West with $\max(24, 2k)$ lags. These regressions use the full sample and are descriptive; Section 8 asks whether anything survives out of sample.

Figure 5 and Table 7 show three facts. First, observed aggressor flow predicts a *reversal*. A one standard deviation increase in taker imbalance is followed by a price decline of 0.7 basis points over the next hour ($t = -3.4$) and 1.3 basis points over two hours ($t = -4.0$); the decline is 1.3 basis points at four hours and has disappeared after about 15 hours. Buying pressure that is absorbed by passive orders is followed by falling prices, which is consistent with a transient component of price impact (Bouchaud et al., 2018) and with passive liquidity providers being, on average, on the right side of the trade.

Second, the information is in the part of flow that the bar's price does not reveal. Holding the standardised bar return fixed leaves the coefficient on observed imbalance essentially unchanged (-1.4 basis points at two hours, $t = -3.7$), while the bar return itself has no predictive power ($t = 0.5$ at two hours and $t = 0.6$ at four). Third, and as a consequence, hourly BVC imbalance predicts nothing: its coefficient is -0.3 basis points at two hours ($t = -0.5$) and is positive, if insignificant, at longer horizons. Section 5 showed that about three-quarters of the variance of observed imbalance is invisible in the hourly bar's price change; this section shows that the invisible part is the part that matters.

Controls and the best available proxy. Table 8 asks whether the reversal is a disguised return or volume effect. Adding three lags of the hourly return, 24-hour momentum, standardised volume,

Table 7: Return response to order-flow shocks, BTC/USDT

Regressor (1 SD)	$k = 1$h	$k = 2$h	$k = 4$h	$k = 8$h	$k = 12$h	$k = 24$h
Observed taker imbalance	−0.72	−1.30	−1.33	−0.97	−0.72	0.57
	(−3.42)	(−3.99)	(−2.58)	(−1.23)	(−0.72)	(0.38)
BVC imbalance	−0.41	−0.30	0.31	1.15	1.40	1.09
	(−1.12)	(−0.55)	(0.42)	(1.17)	(1.16)	(0.56)
Observed imbalance, given z_t	−0.73	−1.41	−1.55	−1.36	−1.37	0.23
	(−2.97)	(−3.74)	(−2.61)	(−1.51)	(−1.20)	(0.13)
Bar return z_t, given imbalance	0.02	0.26	0.50	0.89	1.48	0.78
	(0.06)	(0.46)	(0.64)	(0.84)	(1.15)	(0.38)

Coefficients from regression (2): cumulative log return in basis points from the close of hour t to $t + k$ per one standard deviation of the regressor; Newey-West t-statistics in parentheses. The third and fourth rows come from one regression with both regressors.

volatility and the previous hour's imbalance leaves the two-hour coefficient at −1.3 basis points ($t = -3.7$). The flow effect is not a reversal of past returns and not a volume or volatility effect. The last rows use the most accurate proxy we can build from price data, BVC applied to one-minute bars and summed to the hour, whose correlation with observed imbalance is 0.60 rather than 0.49. On its own it does predict a reversal (−1.0 basis points at two hours, $t = -2.5$). When it competes with observed imbalance, however, its coefficient falls to −0.3 ($t = -0.7$) while observed imbalance keeps −1.1 ($t = -2.6$). Predictive content therefore rises with measurement accuracy: none for hourly BVC, part of the signal for one-minute BVC, and all of it for observed flow, which subsumes the proxy.

Twenty coins. The last columns of Table 3 report the two-hour response for every coin. The coefficient on observed imbalance is negative for 17 of the 20 coins, significantly so at the 5% level for 7, and significantly positive for 1 (TRX, whose flow is followed by continuation); under symmetry a sign pattern this lopsided has a probability of 0.003. Pooling all coins with coin fixed effects and Driscoll and Kraay (1998) standard errors, which allow for any correlation across coins and over time, gives −0.65 basis points per standard deviation ($t = -3.0$), and −0.61 ($t = -2.8$) without bitcoin. The reversal is a market-wide phenomenon. When each coin's imbalance is split into the average imbalance of all coins in that hour and the coin's deviation from it, a one standard deviation increase in market-wide imbalance is followed by a change of −1.50 basis points over two hours ($t = -3.1$), whereas the coin-specific component has a coefficient of 0.01 ($t = 0.1$). Hours in which aggressive buying is broad-based are followed by a partial reversal across the market, consistent with liquidity providers absorbing correlated flow at a temporary price concession; flow specific to one coin carries no comparable signal. In the single-predictor out-of-sample test of Section 8, observed imbalance is significant at the 5% level for 7 of the 20 coins and hourly BVC imbalance for 1.

The magnitude is small. At four hours the observed-imbalance regression explains 0.016% of the return variance, and the implied move of one to two basis points per standard deviation compares with a standard spot fee of ten basis points per side. Even a perfect measurement of flow at this frequency would therefore be of little direct use for trading, a conclusion the next section confirms out of sample.

Table 8: Return response to observed flow with controls and against the best proxy

Specification (coefficient of interest)	$k = 1$h	$k = 2$h	$k = 4$h
BTC/USDT			
(1) taker	−0.72 (−3.4)	−1.30 (−4.0)	−1.33 (−2.6)
(2) + bar return	−0.73 (−3.0)	−1.41 (−3.7)	−1.55 (−2.6)
(3) + lagged returns, 24h momentum	−0.73 (−2.9)	−1.39 (−3.6)	−1.52 (−2.5)
(4) + volume, volatility, lagged taker	−0.64 (−2.5)	−1.32 (−3.7)	−1.52 (−2.8)
(5) 1m BVC alone	−0.65 (−2.4)	−1.00 (−2.5)	−0.75 (−1.3)
(6) taker vs 1m BVC: taker	−0.52 (−1.9)	−1.09 (−2.6)	−1.38 (−2.1)
(6) taker vs 1m BVC: 1m BVC	−0.33 (−1.0)	−0.35 (−0.7)	0.09 (0.1)
ETH/USDT			
(1) taker	0.10 (0.3)	−0.53 (−1.1)	0.55 (0.8)
(2) + bar return	−0.15 (−0.4)	−0.79 (−1.5)	0.21 (0.3)
(3) + lagged returns, 24h momentum	−0.13 (−0.4)	−0.75 (−1.4)	0.24 (0.3)
(4) + volume, volatility, lagged taker	0.00 (0.0)	−0.82 (−1.7)	0.08 (0.1)
(5) 1m BVC alone	0.33 (0.9)	−0.23 (−0.4)	0.24 (0.3)
(6) taker vs 1m BVC: taker	−0.12 (−0.4)	−0.59 (−1.1)	0.60 (0.7)
(6) taker vs 1m BVC: 1m BVC	0.40 (0.9)	0.10 (0.2)	−0.09 (−0.1)

Coefficient (basis points per standard deviation) of observed taker imbalance, or of one-minute BVC imbalance where indicated, in regressions of the k-hour return on standardised regressors; Newey-West t-statistics in parentheses. (2) adds the standardised bar return; (3) adds three lagged hourly returns and 24-hour momentum; (4) adds standardised volume, Garman-Klass volatility and the previous hour's imbalance; (5) uses one-minute BVC imbalance alone; (6) includes observed and one-minute BVC imbalance together.

8 Out-of-sample forecasting and economic value

8.1 Design

Target and walk-forward. The target is the four-hour log return $y_t = \log C_{t+4} - \log C_t$, formed at the close of hour t. The first six months after the feature warm-up are used only for training. The remaining sample (2022-07-10 to 2026-10-31, 37,593 hours) is cut into 53 consecutive 30-day test blocks. Before each block every model is re-estimated on all observations whose label is observable at the block start, i.e. with $t + 4$ strictly before the block (purging); nothing from the block enters estimation. Within each training window the last 20% serves as a validation block, itself purged from the inner training rows, on which hyper-parameters are chosen before the model is refitted on the full window. Winsorisation bounds (1st and 99th percentiles) and standardisation moments are estimated on the training window only. Automated tests replace every bar after t by noise and verify that features at t are unchanged, and check the purge condition for every fold (Appendix B).

Models. Following Gu et al. (2020), models forecast the return relative to zero. The elastic net is fitted without intercept on standardised features, with the penalty chosen from a 13-point grid and the ℓ_1 share from $\{0.1, 0.5, 0.9\}$ on the validation block. As a non-linear benchmark we use gradient-boosted trees with depth three, at least 300 observations per leaf and a learning rate of

Table 9: Out-of-sample forecast accuracy, four-hour BTC/USDT returns

Model	R^2_{OOS} vs. 0 (%)	R^2_{OOS} vs. mean (%)	CW t vs. 0	IC	95% CI	Hit rate	Hit t
Historical mean	−0.075	0.000	–	−0.002	[−0.021, 0.015]	–	–
EN: TI	−0.055	0.020	1.40*	0.002	[−0.014, 0.019]	0.491	−2.14
EN: MS_{proxy}	−0.139	−0.064	−0.61	−0.005	[−0.024, 0.013]	0.488	−2.05
EN: MS	−0.006	0.070	0.86	−0.002	[−0.019, 0.014]	0.494	−0.92
EN: ALL	−0.025	0.050	1.34*	0.019	[0.004, 0.035]	0.508	1.44
EN: Flow_{BVC}	−0.081	−0.005	−1.35	−0.021	[−0.038, −0.006]	0.475	−3.99
EN: $\text{Flow}_{\text{taker}}$	−0.036	0.040	−0.41	−0.007	[−0.021, 0.009]	0.492	−1.26
GBM: MS	−0.284	−0.208	0.39	0.000	[−0.018, 0.019]	0.498	−0.66
GBM: ALL	−0.262	−0.187	0.98	0.003	[−0.014, 0.020]	0.494	−1.72

Monthly expanding-window walk-forward, 2022-07-10 to 2026-10-31 (37,593 hourly forecasts). EN: elastic net; GBM: gradient-boosted trees. CW: Clark-West statistic against the zero forecast (one-sided). IC: Spearman correlation of forecast and outcome, with 95% block-bootstrap interval. Hit t: HAC t-statistic of the hit rate against 50%. *, **, ***: 10%, 5%, 1% significance.

0.03, with the number of rounds chosen on the validation block. We also report the historical-mean forecast. We estimate every model on every feature set defined in Section 4 and report all of them.

Statistical evaluation. The out-of-sample R^2 relative to a benchmark b is $R^2_{OOS} = 1 - \sum(y - \hat{y})^2 / \sum(y - \hat{y}^b)^2$ (Campbell and Thompson, 2008). Because the zero forecast and smaller feature sets are nested in larger ones, we test equal predictive accuracy with the Clark and West (2007) statistic (one-sided), which adjusts for the noise that estimating irrelevant parameters adds to the larger model, and report the Diebold and Mariano (1995) statistic, which compares realised accuracy. Overlapping four-hour targets make forecast errors serially correlated, so all t-statistics use Newey and West (1987) standard errors with $\max(24, 2h)$ lags. The information coefficient (rank correlation of forecast and outcome) is reported with a 95% moving-block bootstrap interval with 48-hour blocks (Künsch, 1989).

Economic evaluation. Each hour a signal $s_t \in \{-1, 0, 1\}$ is formed and held for four hours with a quarter of capital per sleeve (Jegadeesh and Titman, 1993), so the position over $(t, t+1]$ is $\frac{1}{4}\sum_{k=0}^{3} s_{t-k}$; every forecast is used and no exposure is double counted. The *sign* rule sets $s_t = \text{sign}(\hat{y}_t)$, which is zero when the model predicts exactly zero; the *selective* rule trades only when $|\hat{y}_t|$ exceeds the 80th percentile of the model's absolute in-sample forecasts in the current training window. Costs of $c \in \{0, 2, 5, 10\}$ basis points are charged per unit change in position. We report the break-even cost at which net profit is zero, the HAC t-statistic of the intercept in a regression of strategy returns on BTC returns, the number of calendar years with a positive Sharpe ratio, and the deflated Sharpe ratio (DSR) of Bailey and López de Prado (2014). The DSR is the probability that a strategy's true Sharpe ratio exceeds the maximum expected among all 16 model and rule combinations under the null of no skill, given their dispersion, the sample length and the non-normality of returns.

8.2 Forecast accuracy

Table 9 reports the main results for BTC/USDT. All 8 models have a negative out-of-sample R^2 against the zero forecast; the best is EN: MS at −0.006%. The historical mean is also worse than

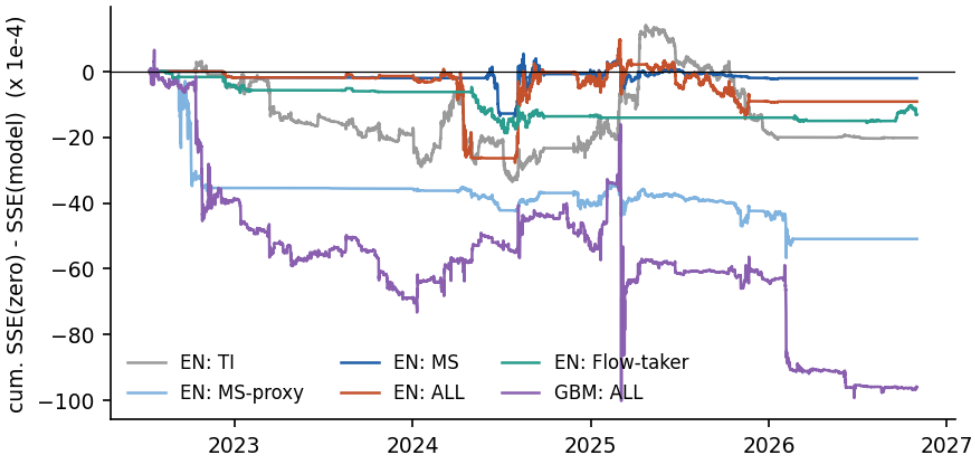


Figure 6: Cumulative squared-error difference between the zero forecast and each model, BTC/USDT four-hour returns. A rising line means the model is more accurate than the zero forecast.

Table 10: Incremental predictive information, BTC/USDT

Model	Benchmark	Nested	R^2_{OOS} rel. (%)	CW t	CW p	DM t	DM p
EN: ALL	EN: TI	yes	0.030	2.41***	0.008	0.33	0.741
EN: MS	EN: TI	yes	0.050	2.85***	0.002	0.55	0.585
EN: MS	EN: MS_{proxy}	yes	0.133	2.59***	0.005	1.53	0.125
EN: MS_{proxy}	EN: TI	yes	−0.084	1.49*	0.068	−0.71	0.476
GBM: ALL	EN: TI	yes	−0.207	1.73**	0.042	−0.88	0.378
EN: $\text{Flow}_{\text{taker}}$	EN: Flow_{BVC}	no	0.045	–	–	0.97	0.332

R^2_{OOS} rel.: out-of-sample R^2 of the model relative to the benchmark. CW: Clark-West test of the null that the larger model adds nothing to the nested benchmark (one-sided). DM: Diebold-Mariano test of equal mean squared error (two-sided). The observed-flow and BVC-flow models are not nested, so only DM is reported.

zero: trailing average returns are a noisy estimate of the drift, and in this sample the early training windows are dominated by the 2022 bear market, which the following recovery reversed (Campbell and Thompson, 2008). Clark-West statistics against zero are significant at the 10% level only for the technical-indicator model ($p = 0.080$) and the full model ($p = 0.091$). The information coefficients are all within about two hundredths of zero. The confidence interval excludes zero for the full model (IC = 0.019) and, with the wrong sign, for the BVC-flow model (IC = −0.021), whose coefficients are unstable across folds. The observed-flow model, which uses the matched constructions from exchange-reported taker volume, does no better than its BVC counterpart.

Figure 6 plots the cumulative difference in squared errors between the zero forecast and each model (Welch and Goyal, 2008). No line trends upwards for an extended period; gains in some months are offset by losses in others.

8.3 Incremental information

Table 10 asks whether each richer feature set improves on a simpler one. The Clark-West test rejects the null that microstructure variables add nothing to technical indicators ($p = 0.002$ for MS over TI

Table 11: Single-predictor out-of-sample test

						R^2_{OOS} (CW p) by half	
Predictor	Horizon	R^2_{OOS} (%)	CW t	CW p	IC	first	second
BTC/USDT							
Observed taker imbalance	1h	0.012	3.19***	0.0007	0.048	0.009 (0.032)	0.016 (0.005)
Observed taker imbalance	2h	0.021	3.80***	0.0001	0.044	0.007 (0.022)	0.038 (0.001)
Observed taker imbalance	4h	0.016	2.62***	0.0044	0.033	0.003 (0.233)	0.032 (0.006)
BVC imbalance (1h bars)	1h	−0.001	0.74	0.2286	0.030	0.007 (0.183)	−0.011 (0.370)
BVC imbalance (1h bars)	2h	−0.007	−0.56	0.7119	0.015	−0.010 (0.848)	−0.004 (0.489)
BVC imbalance (1h bars)	4h	−0.008	−0.35	0.6353	−0.034	−0.007 (0.484)	−0.010 (0.836)
BVC imbalance (1m bars)	1h	0.014	2.39***	0.0085	0.060	0.027 (0.015)	0.000 (0.111)
BVC imbalance (1m bars)	2h	0.017	2.55***	0.0053	0.054	0.023 (0.020)	0.009 (0.062)
BVC imbalance (1m bars)	4h	0.002	1.04	0.1498	0.038	−0.001 (0.319)	0.005 (0.170)
ETH/USDT							
Observed taker imbalance	1h	−0.013	0.63	0.2652	−0.035	−0.025 (0.227)	−0.003 (0.911)
Observed taker imbalance	2h	−0.006	0.14	0.4443	0.002	−0.018 (0.616)	0.004 (0.198)
Observed taker imbalance	4h	−0.015	−0.69	0.7560	−0.016	−0.031 (0.772)	−0.002 (0.499)
BVC imbalance (1h bars)	1h	−0.006	0.42	0.3383	−0.040	−0.023 (0.448)	0.008 (0.127)
BVC imbalance (1h bars)	2h	−0.004	0.41	0.3405	−0.043	−0.005 (0.358)	−0.004 (0.423)
BVC imbalance (1h bars)	4h	0.009	1.20	0.1151	−0.026	−0.011 (0.620)	0.025 (0.063)
BVC imbalance (1m bars)	1h	−0.005	−0.12	0.5462	−0.018	−0.011 (0.612)	0.001 (0.373)
BVC imbalance (1m bars)	2h	−0.005	−0.54	0.7052	0.023	−0.007 (0.733)	−0.003 (0.562)
BVC imbalance (1m bars)	4h	−0.006	−1.19	0.8829	0.013	−0.010 (0.850)	−0.004 (0.725)

One-variable expanding-window regression, re-estimated monthly with purged labels; the coefficient is learned from past data only. R^2_{OOS} against the zero forecast; CW: Clark-West test against zero (one-sided); IC: Spearman correlation of forecast and outcome. The last two columns split the out-of-sample period into halves.

and $p = 0.008$ for ALL over TI) and that observed flow and intrahour measures add nothing to the OHLCV proxies ($p = 0.005$). In each case, however, the realised gain in mean squared error is a few hundredths of a percent and the Diebold-Mariano test does not reject equal accuracy. This is the pattern Clark-West is designed to detect: the larger model's population coefficients are not all zero, but the information they carry is smaller than the noise introduced by estimating them, so the larger model does not forecast better in practice.

The model selection inside each training window tells the same story. For the elastic net the validation step chooses the penalty that sets every coefficient to zero in 57% of the 53 folds with the OHLCV proxies, 68% with observed flow and intrahour measures, and 58% with all features, against 25% with technical indicators alone. Most of the time, the data prefer no forecast at all.

8.4 A single predictor: observed flow against its proxy

The flow models above combine five constructions and tune a penalty on a short validation block, which may discard a weak but genuine signal. Section 7 suggests a direct test: does the current hour's observed taker imbalance, on its own, forecast returns out of sample, and does its BVC counterpart? For each predictor we estimate a one-variable regression without intercept on the standardised predictor in the same expanding monthly walk-forward, with purged labels and moments from the training window only, and no tuning.

Table 11 gives a clean contrast for BTC. Observed taker imbalance has a positive out-of-sample

Table 12: Trading on the single observed-flow forecast, two-hour holding period

		Ann. ret. (%)	Sharpe at cost (bps)			Turnover	Break-even
Asset	Rule	gross	0	1	2	per day	(bps)
BTC/USDT	sign	30.0	0.87	−0.34	−1.54	11.4	0.72
BTC/USDT	selective	26.4	1.43	0.23	−0.97	6.1	1.19
ETH/USDT	sign	10.6	0.23	−0.67	−1.56	11.3	0.26
ETH/USDT	selective	5.9	0.26	−0.62	−1.50	5.4	0.30

Overlapping-sleeve rule on the out-of-sample forecast from observed taker imbalance alone (Table 11, two-hour horizon). Sign: trade every hour in the direction of the forecast; selective: only when $|\hat{y}_t|$ exceeds the 80th percentile of in-sample absolute forecasts. Break-even: cost per unit turnover at which net profit is zero.

R^2 at every horizon, 0.021% at two hours with a Clark-West p-value of 0.0001 and an information coefficient of 0.044. The result holds in both halves of the test period (0.007% with $p = 0.022$ in the first and 0.038% with $p = 0.0005$ in the second). Hourly BVC imbalance has a negative out-of-sample R^2 at every horizon and no significant Clark-West statistic. BVC on one-minute bars lies in between, with an out-of-sample R^2 of 0.017% at two hours ($p = 0.005$), in line with the in-sample gradient of Section 7. Because the coefficient is re-estimated from past data each month, the sign of the relation is learned in real time; the predictor itself, however, was suggested by the full-sample analysis, so the test is not pre-registered. Even a Bonferroni correction for all 45 candidate features leaves the two-hour result for observed flow significant at the 1% level. For ETH none of the three predictors forecasts returns, consistent with the weaker in-sample response.

Two lessons follow. First, the information that order flow carries about short-horizon BTC returns is detectable out of sample when flow is measured, partly recoverable from one-minute prices, and absent from hourly bars. Second, the multivariate elastic net that includes the same variable fails to exploit it: with a weak signal and many correlated candidates, cross-validated shrinkage on a short validation block usually selects the empty model. The signal is nonetheless far too small to trade. Table 12 applies the overlapping-sleeve rule to the observed-flow forecast with a two-hour holding period. Gross Sharpe ratios are 0.87 for the sign rule and 1.43 for the selective rule, but the rules turn over their position many times a day, and their break-even costs are 0.7 and 1.2 basis points per unit of turnover. That is an order of magnitude below the standard fee of ten basis points, and below the taker fees of high-volume accounts, which remain at a few basis points; a strategy that must trade immediately on an hourly signal cannot rely on maker rebates. The predictable component is a statistical regularity, not an exploitable strategy.

8.5 Economic value

Table 13 translates the forecasts into trading rules. Before costs, most rules have Sharpe ratios between −1 and 1, and only 1 of the 16 combinations keeps a positive Sharpe ratio at five basis points. One combination stands out: the sign rule applied to the full elastic net, with a gross Sharpe ratio of 1.24, a break-even cost of 9.0 basis points and almost no market exposure (beta −0.01, alpha $t = 2.7$). Three diagnostics argue against reading it as evidence of a profitable signal. Its probabilistic Sharpe ratio against zero is 0.995, but once we account for having selected it from 16 candidates, the deflated Sharpe ratio is 0.70, below conventional thresholds. Its Sharpe ratio ranges

Table 13: Trading performance of forecast-based rules, BTC/USDT

	Ann. ret. (%)	Sharpe at cost (bps)			Turnover	Break-even	α t	Years	DSR
Model	gross	0	2	5	per day	(bps)	(HAC)	$SR > 0$	
Sign rule									
EN: TI	7.8	0.39	−0.19	−1.05	2.61	1.4	0.78	2/5	0.11
EN: MS_{proxy}	−14.1	−0.38	−0.59	−0.90	0.81	−3.7	−0.88	0/5	0.00
EN: MS	−10.9	−0.40	−0.64	−1.00	0.75	−3.3	−0.83	1/5	0.00
EN: ALL	35.5	1.24	0.97	0.55	1.04	9.0	2.68	3/5	0.70
EN: $Flow_{BVC}$	−22.8	−0.84	−1.00	−1.24	0.59	−10.4	−1.54	0/5	0.00
EN: $Flow_{taker}$	−5.5	−0.17	−0.42	−0.80	0.72	−1.3	−0.17	3/5	0.01
GBM: MS	−10.1	−0.05	−0.46	−1.07	2.33	−0.2	0.25	2/5	0.02
GBM: ALL	−10.7	−0.06	−0.38	−0.87	1.85	−0.4	0.46	3/5	0.02
Selective rule (top quintile of |forecast|)									
EN: TI	10.9	0.71	0.34	−0.21	0.83	3.9	1.51	4/5	0.28
EN: MS_{proxy}	−7.8	−0.47	−0.66	−0.93	0.38	−5.1	−1.21	1/5	0.00
EN: MS	0.6	0.11	−0.12	−0.46	0.33	1.0	0.13	3/5	0.03
EN: ALL	6.1	0.47	0.26	−0.05	0.43	4.5	0.96	4/5	0.14
EN: $Flow_{BVC}$	−8.4	−0.59	−0.75	−0.98	0.29	−7.5	−1.02	1/5	0.00
EN: $Flow_{taker}$	−4.9	−0.45	−0.74	−1.17	0.39	−3.2	−0.89	1/5	0.00
GBM: MS	8.9	0.49	0.26	−0.08	0.71	4.3	1.03	2/5	0.15
GBM: ALL	−2.4	0.00	−0.26	−0.65	0.76	0.0	0.14	2/5	0.02
Buy and hold	34.8	0.87							

Overlapping-sleeve implementation with a four-hour holding period. Costs are per unit change in position. Break-even: cost per unit turnover at which net profit is zero. α t: HAC t-statistic of the intercept in a regression of gross strategy returns on BTC returns. Years $SR > 0$: calendar years with positive gross Sharpe ratio. DSR: deflated Sharpe ratio (Bailey and López de Prado, 2014) across all 16 model and rule combinations.

from −1.8 to 1.9 across calendar years and is positive in 3/5 of them. And it is not the model with the best forecast accuracy, so its performance does not reflect a systematic forecasting advantage. No rule has a break-even cost above 9.0 basis points, so a trader paying the standard fee of ten basis points per side would have lost money with every rule.

9 Robustness

Second asset. We repeat the full analysis for ETH/USDT (Appendix Tables 28, 25 and 29). The measurement results are almost identical. Hourly BVC imbalance has a correlation of 0.49 with observed imbalance, the in-sample information ceiling of return-based classifiers is 0.24 on hourly and 0.29 on one-minute bars, Garman-Klass volatility has a correlation of 0.99 with realised volatility, and the Corwin-Schultz estimate has a median of 20.7 basis points against a tick of 0.043. Section 5.1 extends the measurement results to twenty coins.

The forecasting results for ETH differ in one respect. The technical-indicator model is the only one with a positive out-of-sample R^2 (0.102%, Clark-West $p = 0.001$), and adding microstructure variables makes forecasts significantly *worse*: the Diebold-Mariano p-value for ALL against TI is 0.004, with ALL less accurate. No microstructure or flow model beats the zero forecast. The best trading rule is again one that uses no microstructure information, the sign rule on EN: TI with a gross Sharpe ratio of 1.12, and its deflated Sharpe ratio of 0.75 is below conventional thresholds. The

Table 14: Stability of the return response to observed flow

		Observed imbalance		Observed imbalance | z_t	
Sample	Hours	$k = 2$	$k = 4$	$k = 2$	$k = 4$
BTC/USDT					
Full sample	41,977	−1.30 (−4.0)	−1.33 (−2.6)	−1.41 (−3.7)	−1.55 (−2.6)
Standard fees	35,823	−1.37 (−3.9)	−1.40 (−2.5)	−1.51 (−3.7)	−1.59 (−2.5)
Zero-fee window	6,154	−1.53 (−1.7)	−1.98 (−1.3)	−1.93 (−1.8)	−3.20 (−1.7)
Jan 2022 - Jun 2024	21,506	−1.49 (−2.9)	−0.98 (−1.2)	−1.72 (−2.9)	−1.36 (−1.5)
Jul 2024 - Oct 2026	20,471	−1.27 (−2.9)	−1.63 (−2.4)	−1.33 (−2.6)	−1.69 (−2.1)
ETH/USDT					
Full sample	41,977	−0.53 (−1.1)	0.55 (0.8)	−0.79 (−1.5)	0.21 (0.3)
Standard fees	35,823	−0.67 (−1.3)	0.68 (0.9)	−0.87 (−1.6)	0.30 (0.3)
Zero-fee window	6,154	0.43 (0.4)	−0.71 (−0.4)	−0.15 (−0.1)	−0.97 (−0.5)
Jan 2022 - Jun 2024	21,506	−0.44 (−0.7)	0.10 (0.1)	−0.64 (−0.9)	0.23 (0.2)
Jul 2024 - Oct 2026	20,471	−0.63 (−0.9)	0.87 (0.8)	−0.97 (−1.3)	0.03 (0.0)

Coefficients (basis points per standard deviation) and Newey-West t-statistics from regression (2) at horizons of two and four hours, by sub-sample. The last two columns include the standardised bar return z_t as a control.

Table 15: Forecast horizon, BTC/USDT

Horizon	EN: TI	EN: MS_{proxy}	EN: MS	EN: ALL
1h	0.018 (0.01)	−0.025 (0.73)	0.000 (0.06)	−0.004 (0.05)
4h	−0.055 (0.08)	−0.139 (0.73)	−0.006 (0.19)	−0.025 (0.09)
8h	0.109 (0.00)	0.010 (0.15)	0.007 (0.27)	0.034 (0.04)
24h	0.072 (0.03)	−0.572 (0.78)	−0.841 (0.90)	−0.108 (0.32)

Out-of-sample R^2 against the zero forecast (%), Clark-West p-value against zero in parentheses.

reversal after aggressor flow found for BTC does not replicate for ETH at conventional significance levels: the two-hour coefficient is negative in the full sample, under standard fees and in both halves, but about half as large as for BTC, with t-statistics between −0.7 and −1.7, and it is positive during the zero-fee window (Table 14). In ETH, BVC imbalance has a positive coefficient at eight to twelve hours ($t \approx 2.3$, Appendix Table 27); the bar-return coefficient is similar, so this reflects return continuation rather than information in flow.

Stability of the flow response. Table 14 re-estimates regression (2) in sub-samples. For BTC the two-hour reversal is of similar size under standard fees, during the zero-fee window and in both halves of the sample, and remains when the bar return is held fixed. It is therefore not an artefact of the promotion or of one market phase.

Forecast horizon. Table 15 repeats the walk-forward for horizons of 1, 8 and 24 hours. The pattern matches ETH: the technical-indicator model has small positive out-of-sample R^2 at one, eight and 24 hours, the largest being 0.109% at eight hours (Clark-West $p = 0.002$), while the microstructure feature sets never improve on it.

Estimation window. Replacing the expanding window by a six-month rolling window, which adapts faster to changing conditions at the cost of noisier estimates, makes every model worse, with out-of-sample R^2 of about -0.4% and no significant Clark-West statistic (Appendix Table 21). Whatever predictability exists is not a short-lived, local relation that a shorter window would capture.

Feature groups. Dropping one feature group at a time from the full elastic net (Appendix Table 22) leaves the out-of-sample R^2 between about -0.17% and $+0.02\%$. The only positive value arises when the volatility group is removed, and it is too small to matter; no single group is responsible for the negative results.

Conditional predictability. Predictability could be concentrated in particular states, for example when volatility is high or liquidity is low. Appendix Table 23 splits the out-of-sample period by causal volatility and illiquidity terciles, trading session, weekday and fee regime. Out-of-sample R^2 is close to zero in every cell and no split shows consistent gains across models.

10 Discussion and limitations

What the results do and do not show. The forecasting results are negative, but they are informative because the design rules out the usual explanations. The features are not too few or too coarse: observed flow, trade counts and one-minute realised measures are included. The models are not too simple: shallow boosted trees do no better than the elastic net. The evaluation is not too short: it covers 53 months and a full cycle of bear, recovery and bull markets. What the results show is that, at hourly sampling and a four-hour horizon, the predictable component of BTC and ETH returns linked to these variables is a fraction of a basis point per hour, an order of magnitude below the cost of trading on it.

The natural experiment. The fee experiment has one treated unit, and the main threat is a bitcoin-specific shock that coincided with a switch. Three features of the evidence make this unlikely: the breaks occur in the hour of each switch, the restoration reverses what the removal created, and the same balanced-volume signature appears on whichever bitcoin pair is free at the time. The experiment identifies the response of trading to the removal of fees on one exchange, at a time when other venues charged fees; a market-wide fee change could produce different flows. We cannot say who traded the two-sided volume. High-frequency market makers trading with each other, arbitrageurs across the bitcoin pairs, traders building volume for fee tiers and wash traders would all produce balanced flow, and aggressor flags cannot separate them. Our conclusion about the measures does not depend on the answer: whoever traded it, the volume carried no net direction and did not change the price impact of directional flow, yet volume-based measures read it as liquidity.

Limitations. We do not observe the limit order book, so we cannot measure quoted or effective spreads directly or compute order-flow imbalance at the best quotes (Cont et al., 2014), which is

known to have predictive content at horizons of seconds to minutes (Kolm et al., 2023); for the effect of zero fees on spreads and depth we rely on Galati (2024). Our results therefore speak to bar-based and aggregated trade data, not to high-frequency order-book signals. We study one exchange; aggressor flow on Binance is a large but partial view of a fragmented market (Makarov and Schoar, 2020), and cross-venue flow may be more informative. Taker flags identify the aggressor, not the informed party, and in markets with sophisticated liquidity providers the passive side may be better informed, which is one reading of the reversal in Section 7. Finally, the feature sets, models, walk-forward design and trading rules were fixed before estimation, and we report every specification we ran. The flow-response regressions of Section 7, the single-predictor test of Section 8, the twenty-coin panel and the fee experiment were added after the first results pointed to observed imbalance and to the zero-fee period. The flow-response regressions are descriptive; the single-predictor test, although estimated strictly out of sample, should be read with this sequence and the number of candidate features in mind. The fee experiment was designed around switch dates fixed by the exchange, and we report every outcome we computed.

11 Conclusion

Exchange-reported aggressor volume lets us check what bar-based microstructure variables measure. In twenty cryptocurrencies they measure volatility very well and order flow poorly: in nineteen of them no function of the hourly bar's return explains more than about a quarter of aggressor imbalance, and less for thinly traded coins. Volume-based measures also confuse volume with liquidity. When Binance removed trading fees on bitcoin, two-sided volume multiplied, the Amihud ratio and BVC-based Kyle's lambda signalled a market several times more liquid, and the price impact of a dollar of directional order flow barely changed; the effects appeared and disappeared at the hour of each fee switch and followed the promotion across pairs. Finally, the part of the flow that bars miss is the part that carries information about future prices: market-wide aggressor imbalance is followed by a small reversal, which a single-variable model detects out of sample but which does not survive transaction costs.

For researchers, the implications are concrete. BVC-based imbalance, VPIN and Kyle's lambda computed from bars should not be interpreted as measures of order flow when the aggressor side is available; when it is not, BVC should be computed on the finest bars available and recalibrated as its scale drifts. Volume-based liquidity measures are not comparable across fee regimes, exchanges or periods with different incentives to trade for its own sake, a serious concern in cryptocurrency markets with volume-based fee tiers, fee promotions and wash trading. High-low spread estimators on liquid crypto pairs measure volatility. For forecasting, the binding constraint at this horizon is not model complexity but the size of the signal relative to costs. Order-book data at higher frequency, flow measured across venues, and the identity of the traders behind two-sided volume are the natural next steps.

Data and code availability. All data are public and are downloaded by the scripts in the replication package, which also contains the leakage tests and reproduces every table and figure: `https://github.com/a0x0h/what-bars-cannot-see`.

Acknowledgements. This paper extends a course project in Applied Statistical Analysis at Ferdowsi University of Mashhad. All errors are my own.

References

Alberto Abadie, Alexis Diamond, and Jens Hainmueller. Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505, 2010.

Farshid Abdi and Angelo Ranaldo. A simple estimation of bid-ask spreads from daily close, high, and low prices. *Review of Financial Studies*, 30(12):4437–4480, 2017.

Saketh Aleti and Bruce Mizrach. Bitcoin spot and futures market microstructure. *Journal of Futures Markets*, 41(2):194–225, 2021.

Arash Aloosh and Jiasun Li. Direct evidence of bitcoin wash trading. *Management Science*, 70(12): 8875–8921, 2024.

Diego Amaya, Peter Christoffersen, Kris Jacobs, and Aurelio Vasquez. Does realized skewness predict the cross-section of equity returns? *Journal of Financial Economics*, 118(1):135–167, 2015.

Yakov Amihud. Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5(1):31–56, 2002.

Dan Amiram, Evgeny Lyandres, and Daniel Rabetti. Trading volume manipulation and competition among centralized crypto exchanges. *Management Science*, 71(10):8604–8622, 2025.

Torben G. Andersen and Oleg Bondarenko. VPIN and the flash crash. *Journal of Financial Markets*, 17:1–46, 2014.

David H. Bailey and Marcos López de Prado. The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.

Ole E. Barndorff-Nielsen and Neil Shephard. Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1):1–37, 2004.

Jean-Philippe Bouchaud, Julius Bonart, Jonathan Donier, and Martin Gould. *Trades, Quotes and Prices: Financial Markets Under the Microscope*. Cambridge University Press, 2018.

Christian Bozzetto, Imtiaz Sifat, and Narmin Nahidi. Fee structure and order flow informativeness in the cryptocurrency market. In *Book chapter, Springer, doi:10.1007/978-3-032-18109-1_13*. Springer, 2026.

Alexander Brauneis, Roland Mestel, Ryan Riordan, and Erik Theissen. How to measure the liquidity of cryptocurrency markets? *Journal of Banking & Finance*, 124:106041, 2021.

Alexander Brauneis, Roland Mestel, Ryan Riordan, and Erik Theissen. The anatomy of a fee change: Evidence from cryptocurrency markets. *Journal of Empirical Finance*, 67:152–167, 2022.

John Y. Campbell and Samuel B. Thompson. Predicting excess stock returns out of sample: Can anything beat the historical average? *Review of Financial Studies*, 21(4):1509–1531, 2008.

Bidisha Chakrabarty, Roberto Pascual, and Andriy Shkilko. Evaluating trade classification algorithms: Bulk volume classification versus the tick rule and the Lee-Ready algorithm. *Journal of Financial Markets*, 25:52–79, 2015.

Tarun Chordia, Richard Roll, and Avanidhar Subrahmanyam. Order imbalance, liquidity, and market returns. *Journal of Financial Economics*, 65(1):111–130, 2002.

Todd E. Clark and Kenneth D. West. Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1):291–311, 2007.

CoinDesk. Binance extends zero-fee bitcoin trading globally. https://www.coindesk.com/business/2022/07/06/binance-extends-zero-fee-bitcoin-trading-globally, 2022. 6 July 2022.

CoinDesk. Binance curb on zero-fee trading may cost market share, boost TrueUSD stablecoin: Kaiko research head. https://www.coindesk.com/markets/2023/03/22/binance-curb-on-zero-fee-trading-may-cost-market-share-boost-trueusd-stablecoin, 2023. 22 March 2023.

Jean-Edouard Colliard and Thierry Foucault. Trading fees and efficiency in limit order markets. *Review of Financial Studies*, 25(11):3389–3421, 2012.

Jean-Edouard Colliard and Peter Hoffmann. Financial transaction taxes, market composition, and liquidity. *Journal of Finance*, 72(6):2685–2716, 2017.

Lin William Cong, Xi Li, Ke Tang, and Yang Yang. Crypto wash trading. *Management Science*, 69 (11):6427–6454, 2023.

Timothy G. Conley and Christopher R. Taber. Inference with "difference in differences" with a small number of policy changes. *Review of Economics and Statistics*, 93(1):113–125, 2011.

Rama Cont, Arseniy Kukanov, and Sasha Stoikov. The price impact of order book events. *Journal of Financial Econometrics*, 12(1):47–88, 2014.

Shane A. Corwin and Paul Schultz. A simple way to estimate bid-ask spreads from daily high and low prices. *Journal of Finance*, 67(2):719–760, 2012.

crypto.news. BTC/TUSD volume tumbles 88% as Binance removes zero fees. https://crypto.news/btc-tusd-volume-tumbles-88-as-binance-removes-zero-fees/, 2023. 12 September 2023.

Decrypt. After bitcoin wash trading surges on Binance, CEO nixes incentives. https://decrypt.co/104765/after-bitcoin-wash-trading-surges-on-binance-ceo-nixes-incentives, 2022. 8 July 2022.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

John C. Driscoll and Aart C. Kraay. Consistent covariance matrix estimation with spatially dependent panel data. *Review of Economics and Statistics*, 80(4):549–560, 1998.

David Easley, Marcos López de Prado, and Maureen O'Hara. Flow toxicity and liquidity in a high-frequency world. *Review of Financial Studies*, 25(5):1457–1493, 2012.

David Easley, Marcos López de Prado, and Maureen O'Hara. Discerning information from trade data. *Journal of Financial Economics*, 120(2):269–285, 2016.

David Easley, Marcos López de Prado, Maureen O'Hara, and Zhibai Zhang. Microstructure in the machine age. *Review of Financial Studies*, 34(7):3316–3363, 2021.

David Easley, Maureen O'Hara, Songshan Yang, and Zhibai Zhang. Microstructure and market dynamics in crypto markets. Working paper, SSRN 4814346, 2024.

Thierry Foucault, Ohad Kadan, and Eugene Kandel. Liquidity cycles and make/take fees in electronic markets. *Journal of Finance*, 68(1):299–341, 2013.

Luca Galati. Exchange market share, market makers, and murky behavior: The impact of no-fee trading on cryptocurrency market quality. *Journal of Banking & Finance*, 165:107222, 2024.

Mark B. Garman and Michael J. Klass. On the estimation of security price volatilities from historical data. *Journal of Business*, 53(1):67–78, 1980.

Lawrence R. Glosten and Paul R. Milgrom. Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1):71–100, 1985.

Ruslan Y. Goyenko, Craig W. Holden, and Charles A. Trzcinka. Do liquidity measures measure liquidity? *Journal of Financial Economics*, 92(2):153–181, 2009.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5):2223–2273, 2020.

Campbell R. Harvey, Yan Liu, and Heqing Zhu. . . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68, 2016.

Mohammad R. Jahan-Parvar and Filip Zikes. When do low-frequency measures really measure effective spreads? Evidence from equity and foreign exchange markets. *Review of Financial Studies*, 36(10):4190–4232, 2023.

Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1):65–91, 1993.

Petter N. Kolm, Jeremy Turiel, and Nicholas Westray. Deep order flow imbalance: Extracting alpha at multiple horizons from the limit order book. *Mathematical Finance*, 33(4):1044–1081, 2023.

Hans R. Künsch. The jackknife and the bootstrap for general stationary observations. *Annals of Statistics*, 17(3):1217–1241, 1989.

Albert S. Kyle. Continuous auctions and insider trading. *Econometrica*, 53(6):1315–1335, 1985.

Charles M. C. Lee and Mark J. Ready. Inferring trade direction from intraday data. *Journal of Finance*, 46(2):733–746, 1991.

Yukun Liu and Aleh Tsyvinski. Risks and returns of cryptocurrency. *Review of Financial Studies*, 34(6):2689–2727, 2021.

Igor Makarov and Antoinette Schoar. Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, 135(2):293–319, 2020.

Katya Malinova and Andreas Park. Subsidizing liquidity: The impact of make/take fees on market quality. *Journal of Finance*, 70(2):509–536, 2015.

Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.

Marios A. Panayides, Thomas D. Shohfi, and Jared D. Smith. Bulk volume classification and information detection. *Journal of Banking & Finance*, 103:113–129, 2019.

L. C. G. Rogers and S. E. Satchell. Estimating variance from high, low and closing prices. *Annals of Applied Probability*, 1(4):504–512, 1991.

Richard Roll. A simple implicit measure of the effective bid-ask spread in an efficient market. *Journal of Finance*, 39(4):1127–1139, 1984.

Ivo Welch and Amit Goyal. A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4):1455–1508, 2008.

Dennis Yang and Qiang Zhang. Drift-independent volatility estimation based on high, low, open, and close prices. *Journal of Business*, 73(3):477–491, 2000.

A Feature definitions

Table 16: Feature definitions

Feature	Set	Definition (bar t; r log return, V volume, B taker-buy volume, C, H, L, O prices)
Spread and price impact		
cs_spread_24	MS_{proxy}	24h mean of the two-bar Corwin and Schultz (2012) estimator
ar_spread_24	MS_{proxy}	$\sqrt{\max(0, \overline{4(c_{t-1}-\eta_{t-1})(c_{t-1}-\eta_t)})}$ over 24h, $\eta = (\log H + \log L)/2$ (Abdi and Ranaldo, 2017)
roll_spread_48	MS_{proxy}	$2\sqrt{\max(0, -\operatorname{cov}(r_t, r_{t-1}))}$ over 48h (Roll, 1984)
amihud_24	MS_{proxy}	log of 24h mean of $\lvert r_t\rvert/(C_t V_t)$ (Amihud, 2002)
sqrt_impact_24	MS_{proxy}	log of 24h mean of $\lvert r_t\rvert/\sqrt{V_t/\bar{V}_{168}}$
Order flow, BVC proxy (signed volume $S_t = (2\Phi(r_t/\hat{\sigma}_{168}) - 1)V_t$)		
ofi_1, ofi_24	MS_{proxy}, Flow_{BVC}	$S_t/\bar{V}_{168}$; $\sum_{24} S/\sum_{24} V$
ofi_accel	MS_{proxy}, Flow_{BVC}	6h imbalance minus 24h imbalance
vpin_24	MS_{proxy}, Flow_{BVC}	$\sum_{24} \lvert S\rvert/\sum_{24} V$
kyle_lambda_48	MS_{proxy}, Flow_{BVC}	$\operatorname{cov}(r, SC)/\operatorname{var}(SC)$ over 48h
Order flow, observed (signed volume $2B_t - V_t$) and trade activity		
taker_imb_1, _24, _accel, taker_vpin_24, kyle_lambda_true_48	MS, $\text{Flow}_{\text{taker}}$	as the BVC versions, with observed signed volume
taker_imb_late15	MS	taker imbalance in minutes 45–59 of the hour
trades_z_168, trade_size_z_168	MS	168h z-scores of log trade count and log average dollar trade size
Volatility		
gk_vol_24, rs_vol_24, yz_vol_24	MS_{proxy}	log 24h Garman and Klass (1980), Rogers and Satchell (1991), Yang and Zhang (2000) volatility
vol_ratio_24_168	MS_{proxy}	log ratio of 24h to 168h Garman-Klass volatility
vol_of_vol_168	MS_{proxy}	168h standard deviation of log 24h Garman-Klass volatility
rv_1m_1, rv_1m_24	MS	log realised variance from 1-minute returns, current hour and 24h
jump_share_24	MS	24h mean of $\max(RV - BV, 0)/RV$ (Barndorff-Nielsen and Shephard, 2004)
rskew_24	MS	24h mean of hourly realised skewness (Amaya et al., 2015)
Efficiency, bar structure, activity		
eff_ratio_24	MS_{proxy}	$\lvert\log C_t - \log C_{t-24}\rvert/\sum_{24} \lvert r\rvert$
autocorr_48	MS_{proxy}	48h first-order autocorrelation of r
var_ratio_4_168	MS_{proxy}	variance of 4h returns over 4 times variance of 1h returns, 168h
body_share, upper_wick, lower_wick (24h)	MS_{proxy}	candle body and wicks as share of the high-low range
volume_z_168, volume_trend_24_168	MS_{proxy}	z-score of log volume; log ratio of 24h to 168h mean volume
Technical controls		
mom_1, _4, _24, _168	TI	$\log C_t - \log C_{t-k}$
rsi_14, macd_hist, bb_pctb_20	TI	RSI(14); MACD(12,26,9) histogram over price; Bollinger %B(20,2)
ma_gap_24, ma_gap_168, atr_14	TI	log gap to moving average; ATR(14) over price

B Leakage controls and a positive control

The replication package contains automated tests that run in a few seconds. (i) *Causality of features*: for three cut-off dates, every bar after the cut-off is multiplied by random noise and every feature at or before the cut-off is recomputed; the test fails if any value changes. The same test is applied to the intrahour features built from one-minute bars. (ii) *Target definition*: y_t must equal $\log C_{t+h} - \log C_t$ exactly. (iii) *Purging*: for every fold and every horizon, the last training label must be observed before the first test bar, and the same must hold for the inner validation split. (iv) *Analyses*: every validation and robustness function is run on synthetic data and checked for internal consistency (for example, the in-sample ceiling must weakly exceed the squared correlation of BVC), a pure-noise predictor must not produce an out-of-sample gain in the single-predictor walk-forward, and a planted flow signal must be recovered from observed flow but not from technical momentum.

As a positive control we run the full pipeline on a synthetic two-year hourly series with GARCH-type volatility, in which a persistent latent order flow drives the taker share of volume and each hour's return loads negatively on the average flow of the previous four hours. The contemporaneous link between flow and the bar's return is deliberately weak, so BVC recovers the flow only poorly, as in the real data. The pipeline should find the signal with observed-flow features and much less of it with BVC features, and nothing with technical indicators. It does: the observed-flow elastic net reaches an out-of-sample R^2 of about 3% with a Clark-West t-statistic above 10, the BVC-flow model about zero, and the technical model a negative R^2. A pipeline that leaked future information would instead show gains for every feature set. The script `scripts/make_synthetic.py` generates the series.

C Zero-fee experiment: robustness

Table 17: Effects of removing and restoring fees: alternative specifications

	Event windows				Full windows	
Measure	Main (90 days)	ETH only	60 days	180 days	Means	With trend
Panel A. Fees removed (8 July 2022)						
Trades per day (millions)	1.68 (12.2)	1.25 (11.0)	1.53 (10.1)	2.04 (14.7)	2.14 (16.0)	1.58 (9.7)
Dollar volume per day ($bn)	1.21 (5.9)	1.03 (4.4)	0.90 (6.6)	1.62 (9.5)	1.77 (10.5)	1.09 (4.8)
Average trade size ($)	−0.46 (−4.3)	−0.22 (−1.4)	−0.63 (−10.4)	−0.41 (−6.5)	−0.37 (−6.7)	−0.49 (−4.6)
Directional volume $\sum_m \lvert B_m - S_m \rvert$ ($bn)	0.43 (3.3)	0.25 (1.4)	0.21 (2.6)	0.72 (5.7)	0.85 (6.4)	0.29 (2.0)
Two-sided volume ($bn)	1.43 (6.0)	1.22 (4.9)	1.09 (6.1)	1.91 (9.5)	2.04 (11.1)	1.34 (4.8)
Two-sided share of volume	0.16 (6.2)	0.15 (8.6)	0.14 (4.7)	0.20 (9.9)	0.19 (12.0)	0.18 (4.7)
Mean |hourly taker imbalance|	−1.03 (−9.3)	−0.86 (−7.3)	−0.98 (−6.9)	−1.22 (−12.8)	−1.25 (−16.1)	−1.07 (−6.2)
Amihud ratio, hourly bars ($\lvert r \rvert$ per $bn)	−1.18 (−6.9)	−1.20 (−5.8)	−0.93 (−7.0)	−1.81 (−8.6)	−1.96 (−10.3)	−0.98 (−5.1)
Kyle's λ, BVC-signed flow (bps per $M)	−0.39 (−3.1)	−0.52 (−3.0)	−0.22 (−2.2)	−0.99 (−5.2)	−1.14 (−6.3)	−0.29 (−1.7)
VPIN, BVC	−0.10 (−4.7)	−0.08 (−3.5)	−0.08 (−3.7)	−0.11 (−5.7)	−0.11 (−6.4)	−0.06 (−2.0)
Kyle's λ, observed flow (bps per $M)	0.24 (1.4)	−0.03 (−0.2)	0.35 (1.8)	−0.22 (−1.2)	−0.29 (−1.7)	0.30 (1.4)
VPIN, observed flow	−1.08 (−8.8)	−0.90 (−7.2)	−1.01 (−6.9)	−1.26 (−12.2)	−1.29 (−15.4)	−1.11 (−6.0)
R^2 of 1m returns on observed flow	0.13 (5.2)	0.08 (3.1)	0.11 (4.5)	0.14 (7.5)	0.17 (8.1)	0.10 (3.1)
corr(BVC, observed imbalance), 1m	0.05 (4.1)	0.00 (0.2)	0.04 (2.7)	0.08 (6.2)	0.10 (6.6)	0.03 (1.6)
Realised volatility (% per day)	0.15 (1.4)	−0.28 (−2.4)	0.11 (0.9)	−0.26 (−1.6)	−0.22 (−1.6)	0.25 (1.3)
First-order autocorrelation of 1m returns	−0.01 (−0.5)	−0.01 (−1.7)	−0.02 (−2.5)	0.01 (1.2)	0.00 (0.4)	0.00 (−0.2)
Panel B. Fees restored (22 March 2023)						
Trades per day (millions)	−1.68 (−18.1)	−1.84 (−31.7)	−1.70 (−16.0)	−1.71 (−16.2)	−1.59 (−13.5)	−2.12 (−12.6)
Dollar volume per day ($bn)	−1.18 (−11.2)	−1.56 (−33.1)	−1.26 (−11.7)	−1.26 (−10.9)	−1.09 (−7.7)	−1.74 (−9.0)
Average trade size ($)	0.50 (7.5)	0.28 (4.6)	0.44 (10.3)	0.45 (8.9)	0.50 (9.8)	0.38 (5.4)
Directional volume $\sum_m \lvert B_m - S_m \rvert$ ($bn)	−0.27 (−3.1)	−0.59 (−11.2)	−0.36 (−4.0)	−0.24 (−2.8)	−0.13 (−1.5)	−0.67 (−5.2)
Two-sided volume ($bn)	−1.33 (−12.0)	−1.79 (−30.8)	−1.38 (−11.4)	−1.48 (−10.7)	−1.30 (−7.5)	−1.96 (−8.5)
Two-sided share of volume	−0.15 (−8.9)	−0.18 (−9.5)	−0.13 (−11.4)	−0.19 (−9.2)	−0.18 (−8.4)	−0.19 (−6.1)
Mean |hourly taker imbalance|	1.15 (21.3)	1.18 (19.5)	1.08 (22.3)	1.26 (18.9)	1.16 (13.5)	1.33 (9.8)
Amihud ratio, hourly bars ($\lvert r \rvert$ per $bn)	1.52 (21.8)	1.77 (20.7)	1.49 (38.8)	1.39 (18.2)	1.16 (8.3)	2.09 (11.6)
Kyle's λ, BVC-signed flow (bps per $M)	0.97 (15.2)	1.14 (17.4)	0.98 (13.6)	0.90 (12.0)	0.70 (5.9)	1.51 (10.2)
VPIN, BVC	0.11 (5.1)	0.07 (2.6)	0.10 (4.8)	0.08 (3.4)	0.07 (2.8)	0.11 (4.1)
Kyle's λ, observed flow (bps per $M)	0.30 (4.1)	0.65 (5.9)	0.24 (6.0)	0.19 (2.2)	0.09 (1.0)	0.65 (3.8)
VPIN, observed flow	1.11 (23.1)	1.14 (25.1)	1.04 (31.0)	1.23 (19.6)	1.13 (13.3)	1.29 (9.8)
R^2 of 1m returns on observed flow	−0.06 (−3.8)	−0.03 (−1.3)	−0.08 (−6.8)	−0.04 (−1.7)	−0.01 (−0.7)	−0.08 (−2.6)
corr(BVC, observed imbalance), 1m	−0.05 (−6.2)	−0.04 (−3.3)	−0.04 (−7.8)	−0.03 (−2.2)	−0.01 (−0.4)	−0.07 (−3.5)
Realised volatility (% per day)	0.48 (1.9)	0.31 (2.2)	0.23 (1.4)	0.17 (0.8)	0.06 (0.3)	0.51 (1.5)
First-order autocorrelation of 1m returns	0.02 (2.0)	0.01 (1.2)	0.04 (3.6)	0.02 (1.3)	0.02 (1.6)	0.01 (0.7)

Change in the BTC-minus-control difference, log points for measures in logs; Newey-West t-statistics in parentheses. Main: 19 control coins, 90 days on each side of the switch (Table 4). ETH only: ETH/USDT as the single control. 60 and 180 days: shorter and longer event windows. Full windows: regression of the daily difference on indicators for the zero-fee window (9 July 2022 to 21 March 2023) and the window after it (22 March to 6 September 2023), with 1 January to 7 July 2022 as the base, without and with a linear time trend.

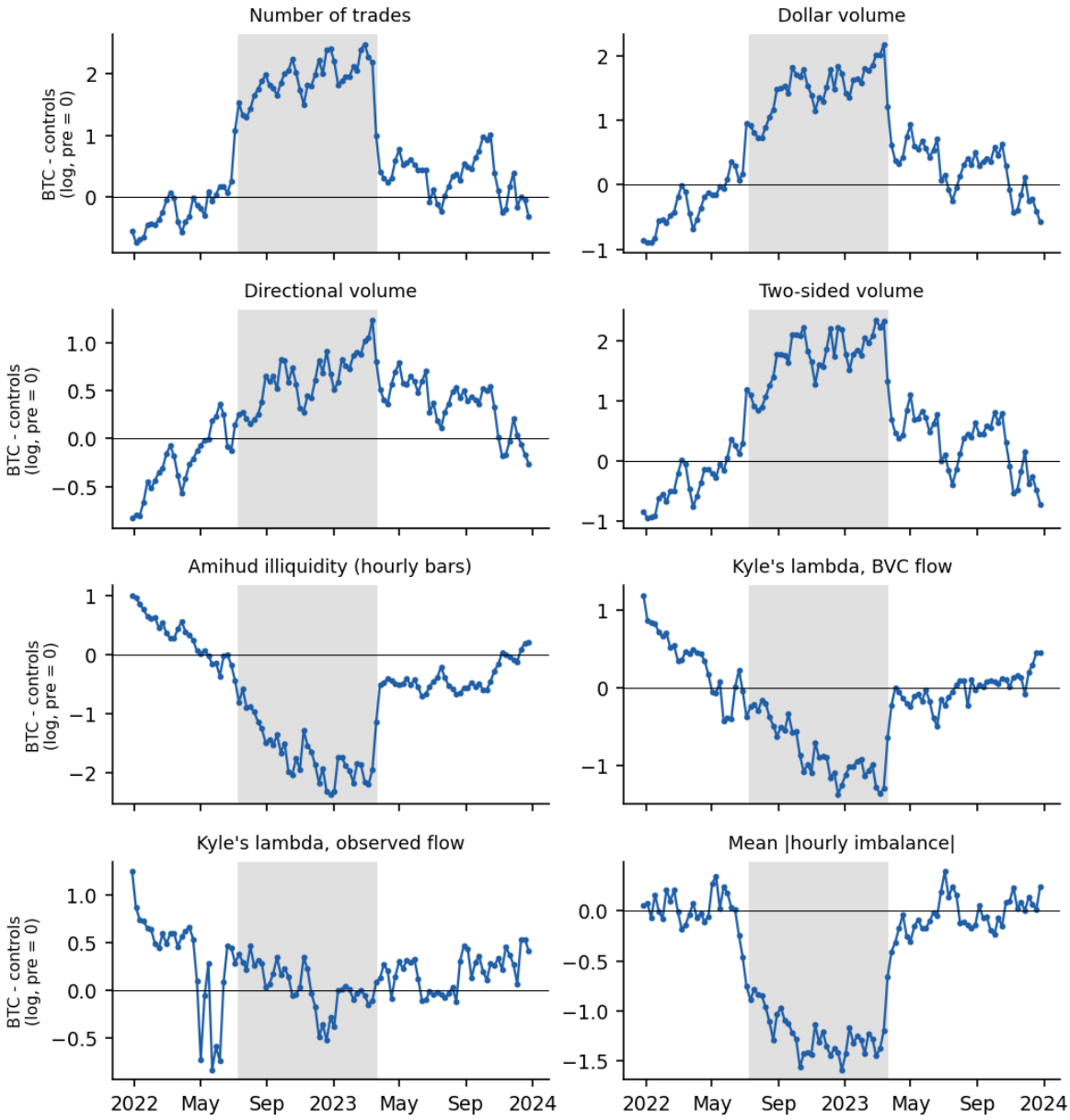


Figure 7: Weekly means of the BTC-minus-control difference for selected daily measures, January 2022 to December 2023, relative to the 90 days before the promotion (log points). Shaded: zero fees on BTC/USDT.

D Additional tables

Table 18: Alternative flow classifiers vs. observed taker imbalance, BTC/USDT

	Pearson		Same sign	
Classifier	1h	24h	1h	24h
BVC Normal, sigma 24h (1h bars)	0.503	0.516	0.716	0.695
BVC Normal, sigma 168h (1h bars)	0.489	0.507	0.716	0.693
BVC Normal, sigma 720h (1h bars)	0.484	0.505	0.716	0.691
BVC Student-t(3), sigma 168h (1h bars)	0.494	0.507	0.716	0.687
Tick rule (1h bars)	0.430	0.456	0.716	0.666
BVC Normal, sigma 1440m (1m bars)	0.603	0.602	0.748	0.725
Tick rule (1m bars)	0.597	0.617	0.728	0.731

Correlation and sign agreement between proxy imbalance and observed taker imbalance, per hour (1h) and over rolling 24-hour sums (24h). Student-$t(3)$ is rescaled to unit variance.

Table 19: Information ceiling by period

Period	Hours	Ceiling	corr^2(BVC)	Slope	SD imb.	Ceiling (1m)
BTC/USDT						
full sample	42,315	0.247	0.239	0.128	0.130	0.338
zero-fee window	6,154	0.374	0.360	0.038	0.030	0.345
standard fees	36,161	0.267	0.257	0.142	0.140	0.362
2022	8,718	0.242	0.234	0.065	0.066	0.248
2023	8,757	0.278	0.253	0.106	0.099	0.370
2024	8,784	0.332	0.315	0.135	0.121	0.411
2025	8,760	0.281	0.265	0.167	0.165	0.362
2026	7,296	0.254	0.246	0.169	0.172	0.347
ETH/USDT						
full sample	42,315	0.244	0.237	0.123	0.126	0.285
zero-fee window	6,154	0.265	0.254	0.098	0.095	0.253
standard fees	36,161	0.244	0.237	0.127	0.131	0.291
2022	8,718	0.283	0.275	0.094	0.091	0.211
2023	8,757	0.246	0.228	0.118	0.119	0.331
2024	8,784	0.319	0.307	0.129	0.118	0.336
2025	8,760	0.250	0.241	0.132	0.136	0.296
2026	7,296	0.212	0.199	0.147	0.163	0.270

Ceiling: in-sample R^2 of observed hourly taker imbalance on 50 quantile-bin means of the standardised bar return, the upper bound for any return-based classifier calibrated to that period. corr^2(BVC): squared correlation of hourly BVC with observed imbalance. Slope: OLS slope of observed imbalance on BVC (its calibration). SD imb.: standard deviation of observed hourly imbalance. Ceiling (1m): the same bound for one-minute imbalance on one-minute returns.

Table 20: BVC accuracy by regime, BTC/USDT

		Pearson		Same sign		
Split	Group	1h	24h	1h	24h	Hours
volatility	high vol	0.499	0.619	0.715	0.744	13,119
volatility	low vol	0.493	0.432	0.717	0.649	15,252
volatility	mid vol	0.503	0.507	0.715	0.690	13,368
year	2022	0.484	0.580	0.721	0.720	8,718
year	2023	0.503	0.497	0.728	0.661	8,758
year	2024	0.561	0.595	0.728	0.701	8,784
year	2025	0.515	0.544	0.707	0.676	8,760
year	2026	0.496	0.533	0.690	0.709	7,296
fee regime	standard fees	0.507	0.524	0.710	0.694	36,162
fee regime	zero-fee promo	0.600	0.591	0.749	0.689	6,154

Volatility terciles use the 24h Garman-Klass volatility ranked against the trailing 30 days.

Table 21: Rolling six-month estimation window, BTC/USDT

Model	Expanding R^2_{OOS}	Rolling R^2_{OOS}	Rolling CW p	Rolling IC
EN: TI	−0.055	−0.411	0.52	0.001
EN: MS_{proxy}	−0.139	−0.379	0.73	−0.007
EN: MS	−0.006	−0.400	0.55	−0.002
EN: ALL	−0.025	−0.433	0.33	−0.006

Out-of-sample R^2 (%) against the zero forecast with the expanding window of the main analysis and with a six-month rolling window; Clark-West p-value and IC for the rolling window.

Table 22: Drop-one-group ablation, full elastic net, BTC/USDT

Group dropped	Features	R^2_{OOS} (%)	IC
none (EN: ALL)	45	−0.025	0.019
spread impact	41	−0.115	−0.004
order flow	40	−0.072	0.014
volatility	39	0.021	0.018
efficiency structure	39	−0.169	0.006
activity	43	−0.045	0.011
trade level	37	−0.125	0.005
intrahour 1m	41	−0.118	0.017

Table 23: Conditional out-of-sample accuracy, BTC/USDT

Split	Group	Hours	EN: TI	EN: MS	EN: ALL	EN: $\text{Flow}_{\text{taker}}$
vol regime	high	11,647	−0.150 (0.38)	−0.029 (0.47)	−0.080 (0.36)	−0.033 (0.72)
vol regime	low	13,914	0.042 (0.05)	−0.056 (0.46)	0.054 (0.06)	−0.087 (0.73)
vol regime	mid	12,032	0.035 (0.05)	0.085 (0.03)	0.006 (0.12)	0.005 (0.25)
illiq regime	high	12,524	−0.049 (0.17)	−0.026 (0.39)	0.056 (0.12)	0.002 (0.32)
illiq regime	low	13,471	0.103 (0.03)	0.070 (0.02)	0.024 (0.08)	−0.043 (0.67)
illiq regime	mid	11,598	−0.254 (0.64)	−0.074 (0.65)	−0.181 (0.55)	−0.072 (0.85)
session	Asia 00-08	12,528	−0.166 (0.33)	−0.086 (0.58)	−0.355 (0.73)	0.002 (0.19)
session	Europe 08-16	12,533	0.014 (0.14)	0.083 (0.03)	0.134 (0.02)	−0.053 (0.83)
session	US 16-24	12,532	−0.077 (0.23)	−0.076 (0.66)	−0.026 (0.23)	−0.038 (0.61)
weekend	weekday	26,867	−0.050 (0.13)	−0.005 (0.24)	0.020 (0.05)	−0.043 (0.80)
weekend	weekend	10,726	−0.085 (0.16)	−0.010 (0.27)	−0.287 (0.52)	0.007 (0.21)
fee regime	standard fees	31,483	−0.032 (0.05)	−0.001 (0.17)	−0.026 (0.08)	−0.026 (0.52)
fee regime	zero-fee (BTC/USDT promo)	6,110	−0.137 (0.64)	−0.023 (0.92)	−0.023 (0.92)	−0.072 (0.84)

Out-of-sample R^2 against zero (%) within each group, Clark-West p-value in parentheses. Volatility and illiquidity terciles rank the 24h Garman-Klass volatility and the Amihud ratio against the trailing 30 days, so group membership is known at forecast time.

Table 24: Sub-period accuracy, BTC/USDT

Sub-period	BTC ret. (%)	EN: TI	EN: MS_{proxy}	EN: MS	EN: ALL	EN: Flow_{BVC}	EN: $\text{Flow}_{\text{taker}}$
2022-07-10–2023-08-14	36	−0.162	−0.354	−0.018	−0.019	−0.228	−0.057
2023-08-14–2024-09-08	84	−0.086	−0.035	0.045	0.034	−0.058	−0.068
2024-09-08–2025-10-05	132	0.319	−0.011	−0.050	−0.079	−0.008	−0.016
2025-10-05–2026-10-31	−38	−0.272	−0.137	−0.009	−0.052	0.000	0.012

Out-of-sample R^2 against zero (%) in four equal sub-periods of the test sample, with the BTC return over each sub-period.

Table 25: Out-of-sample forecast accuracy, ETH/USDT

Model	R^2_{OOS} vs. 0 (%)	R^2_{OOS} vs. mean (%)	CW t vs. 0	IC	95% CI	Hit rate	Hit t
Historical mean	−0.083	0.000	–	−0.009	[−0.028, 0.008]	–	–
EN: TI	0.102	0.186	2.97^{***}	0.012	[−0.003, 0.029]	0.499	−0.20
EN: MS_{proxy}	−0.266	−0.182	−1.60	−0.005	[−0.022, 0.011]	0.495	−0.83
EN: MS	−0.335	−0.252	−1.72	0.003	[−0.015, 0.020]	0.497	−0.46
EN: ALL	−0.356	−0.273	−1.63	−0.007	[−0.025, 0.010]	0.491	−1.41
EN: Flow_{BVC}	−0.083	0.001	−0.34	−0.014	[−0.031, 0.003]	0.491	−1.91
EN: $\text{Flow}_{\text{taker}}$	−0.010	0.073	1.06	0.009	[−0.008, 0.026]	0.505	1.08
GBM: MS	−0.441	−0.357	−0.32	0.002	[−0.016, 0.017]	0.501	0.25
GBM: ALL	−0.478	−0.394	−0.62	−0.016	[−0.034, 0.001]	0.489	−2.93

Table 26: Incremental predictive information, ETH/USDT

Model	Benchmark	Nested	R^2_{OOS} rel. (%)	CW t	CW p	DM t	DM p
EN: ALL	EN: TI	yes	−0.459	−1.09	0.861	−2.90***	0.004
EN: MS	EN: TI	yes	−0.438	−0.87	0.808	−2.64***	0.008
EN: MS	EN: MS_{proxy}	yes	−0.069	0.37	0.354	−0.71	0.478
EN: MS_{proxy}	EN: TI	yes	−0.369	−0.47	0.682	−2.87***	0.004
GBM: ALL	EN: TI	yes	−0.580	−0.36	0.639	−2.70***	0.007
EN: $\text{Flow}_{\text{taker}}$	EN: Flow_{BVC}	no	0.072	–	–	1.30	0.194

Table 27: Return response to order-flow shocks, ETH/USDT

Regressor (1 SD)	$k = 1$h	$k = 2$h	$k = 4$h	$k = 8$h	$k = 12$h	$k = 24$h
Observed taker imbalance	0.10	−0.53	0.55	0.61	0.90	2.68
	(0.33)	(−1.14)	(0.76)	(0.53)	(0.58)	(1.21)
BVC imbalance	0.56	0.81	1.85	3.62	4.58	3.67
	(1.10)	(1.10)	(1.74)	(2.29)	(2.36)	(1.31)
Observed imbalance, given z_t	−0.15	−0.79	0.21	−0.41	−0.46	2.10
	(−0.45)	(−1.55)	(0.26)	(−0.31)	(−0.26)	(0.81)
Bar return z_t, given imbalance	0.58	0.62	0.80	2.39	3.21	1.36
	(1.10)	(0.85)	(0.76)	(1.61)	(1.69)	(0.51)

Table 28: Proxy validation, ETH/USDT

	Pearson	Spearman	Same sign	N
Panel A. Signed order flow: proxy vs. observed taker imbalance				
BVC, 1h bars, per hour	0.487	0.532	0.708	42,316
BVC, 1h bars, 4h sums	0.470	0.530	0.708	42,310
BVC, 1h bars, 24h sums	0.513	0.561	0.720	42,270
BVC, 1m bars summed to 1h	0.556	0.605	0.735	42,359
Tick rule, 1h bars, per hour	0.432	0.418	0.708	42,357
Panel B. Price impact and volatility				
Kyle λ (48h): BVC vs. observed flow	0.440	0.503	–	41,978
log GK vol (1h bars) vs. log RV (1m), 24h	0.987	0.984	–	41,978
Panel C. Information ceiling of return-based classifiers (out-of-sample R^2)				
	BVC raw	BVC rescaled	Any $f(z)$	N
1h bars	−8.868	0.218	0.222	21,158
1m bars	−0.153	0.276	0.286	1,270,544
Panel D. Spread estimators (one tick = 0.043 bps)				
	Median (bps)	Mean (bps)	Share = 0	Corr. log RV
Corwin-Schultz (24h mean)	20.7	23.2	0.0%	0.85
Abdi-Ranaldo (24h)	0.0	11.1	54.7%	0.61
Roll (48h)	17.6	24.0	40.6%	0.52

Table 29: BVC accuracy by regime, ETH/USDT

		Pearson		Same sign		
Split	Group	1h	24h	1h	24h	Hours
volatility	high vol	0.500	0.585	0.714	0.745	13,547
volatility	low vol	0.478	0.468	0.697	0.699	15,050
volatility	mid vol	0.510	0.541	0.715	0.717	13,142
year	2022	0.525	0.666	0.713	0.758	8,718
year	2023	0.478	0.462	0.700	0.678	8,758
year	2024	0.554	0.575	0.727	0.733	8,784
year	2025	0.491	0.541	0.703	0.725	8,760
year	2026	0.446	0.477	0.693	0.701	7,296
fee regime	standard fees	0.487	0.515	0.708	0.723	36,162
fee regime	zero-fee promo	0.504	0.571	0.710	0.700	6,154